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Scientific background document

AI for production scale-up, medical countermeasures distribution and decision-making in crisis preparedness and response

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Executive summary

Health crises significantly impact populations and require urgent and coordinated **resource mobilisation**. Health emergencies often reveal weaknesses in healthcare systems, such as slow processes, lack of essential **medical countermeasures (MCMs)**, inefficient stock management, poor drug traceability, and inadequate data utilisation. In this respect, the EU4Health programme 2021-2027 is paving the way for a strong European Health Union, including among its key priorities reinforcing the European Union's resilience to cross-border health threats.

Following the onset of the COVID-19 pandemic, the European Union (EU) established in 2021 the **European Health Emergency Preparedness and Response** (HERA) authority aimed at contributing to the digital transformation of health systems. HERA plays an essential role in ensuring the **resilience of health systems** across the EU by promoting advanced research and development of MCMs for tackling priority cross-border health threats. Since its creation, HERA has worked with Member States (MS) and various stakeholders, including industry and civil society, to enhance **preparedness and response to health threats**, particularly concerning MCMs. The stockpiling and shortage prevention of MCMs constitute core components of HERA's mandate involving acquiring critical medical items such as medicines and vaccines to maintain a high level of preparedness within the EU against future cross-border health outbreaks.

In recent years, **artificial intelligence (AI)** has significantly contributed to enhancing different aspects of MCMs production, distribution and stockpiling to improve the efficiency of the MCMs supply chain (Gao et al., 2023a; Hu et al., 2023a). Even before the COVID-19 outbreak, the value of AI technologies in preparing for and responding to health emergencies was recognised, with some applications already in use, for example, supporting policymakers and stakeholders in decision-making. However, the global pandemic marked a pivotal moment in the development of AI-based tools. These tools have since advanced in improving outbreak detection, drug development and distribution, predicting MCM shortages, and identifying potential strategies to cope with unexpected events (Faiyazuddin et al., 2024).

This scientific background document aims to contribute to the common EU strategy for stockpiles and supplies of critical MCMs for health emergency preparedness and response. The study identifies the **key areas of current use of AI-based tools across MCMs production, stockpiling and distribution** where AI can be applied to make EU preparedness and response to health emergencies more effective and efficient. It also identifies AI solutions that can contribute to **supporting decision-making** during health crises by identifying and monitoring health threats to implement and prioritise control measures. These tools can also guide experts on MCMs demands and propose strategies to prevent MCMs shortages and accurate distribution and stockpiling. Finally, this report includes the most crucial bottlenecks identified in the implementation of AI-

based tools and describes their current market availability and implementation, ultimately strengthening health emergency and response.

AI-based tools have been identified through a **structured literature review and consultation activities, including in-depth interviews and a survey** addressed to relevant stakeholders from academia, industry and governmental sectors. The results are structured in five functional categories: 1) AI for MCMs production scale-up; 2) AI for MCMs stockpiling and distribution; 3) AI for supporting decision-making; 4) advantages and disadvantages of the identified AI tools, and 5) bottlenecks to the implementation of AI tools in the targeted areas.

AI-based tools to enhance **MCMs production scale-up** are found to support the acceleration of the pipeline from initial research to potential treatments and interventions by expediting the analysis of vast and complex datasets using ML techniques. In addition, AI-based tools are also advancing considerably in cell line engineering by improving the efficiency and scalability of bioprocesses, facilitating the production of personalised medicine and tailored therapies (Gupta, 2023). During health emergencies, AI-driven solutions have been used to automate key processes of MCMs production, such as parameter optimisation for improving production efficiency and enhancing forecast demand based on disease spread. (Gama et al., 2022). This automation has contributed to reducing both the manufacturing time and associated costs, allowing for a more rapid response (Kolluri et al., 2022). Furthermore, blockchain technology can be integrated to ensure the quality and safety of these products The experts also highlighted the importance of incorporating digital twins to optimise MCM process design. This includes testing the outcomes of different drugs and vaccines to identify the most efficient options, defining process ranges, and establishing proven acceptable ranges to ensure consistent performance across different scales.

In **MCMs stockpiling and distribution**, machine learning (ML) models have significantly contributed to the optimisation of MCMs allocation and delivery processes according to real-time demands by leveraging sales and epidemiological surveillance data (Goodarzian et al., 2023). In some cases, blockchain technology has been used in combination with other AI technologies thanks to its capacity for providing a secure data infrastructure. To develop and test ML algorithms, evidence has shown that high-quality and diverse real-time data is essential to ensure the outcomes and solutions offered by the AI tools are reliable. In addition, the use of AI-powered routing algorithms, autonomous vehicles (AVs) and unmanned aerial vehicles (UAVs) represents an opportunity to enhance MCMs distribution by being able to reach remote and limited-capacity regions, broadening MCMs accessibility and relieving the health personnel during health emergencies. Moreover, evidence shows that ML-based tools can contribute to the **early detection of sales anomalies and help detect panic buying and pre-empting hoarding trends,** preventing the “bullwhip effect”, for example. In short, these tools can support MCMs production and distribution according to real-time needs and **optimise resource allocation**.

AI-based tools to support decision-making are based on forecasting future epidemic outbreaks, monitoring the actual spread of disease in regions through AI-driven integrated surveillance systems, and **prioritising the measures to be implemented to face pandemics**. These tools can help European, regional and local authorities identify the best MCMs production, storage and distribution strategies to improve efficiency and rapid response to a health emergency. Evidence and experts suggested knowledge extraction models based on **large language models** improve decision-making in health crises drawing on different data sources, especially if combined with other technologies such as retrieval augmented generation techniques to ensure the safety and quality of the outcomes of the algorithm and be able to inform the population and the final tool's user on the source of the data used to generate results.

Finally, this scientific background document also highlights **the bottlenecks and barriers to implementing AI tools** based on the literature review and the consultation activities with experts. According to the study results, the prioritised bottlenecks to be addressed are **data availability, quality and interoperability issues, as well as the lack of standardised databases and validation procedures**. Other important barriers are the need to offer solutions for the high computational requirements of these AI-based tools, the regulatory hurdles and uncertainties, the **user acceptability** and the **ethical and security challenges**. Indeed, to fully benefit from AI's potential to enhance the healthcare supply chain, it is essential to overcome the barriers and bottlenecks associated with its development and implementation and to adopt procedures, standards (e.g. gold standards for validation) and frameworks (e.g. regulatory) that ensure the safe deployment of AI-based tools in real-world environments.

Another interesting point raised by this study is the need to **improve collaboration** between healthcare organisations, industry and policymakers to improve the response to MCMs demands, prevent shortages and improve overall MCMs supply efficiency. The collaboration between stakeholders should aim at sharing and using real-time data from suppliers, inventory levels, production schedules, and customer demand. In addition, the implementation of **collaborative networks** between stakeholders is crucial to **successfully integrate AI** into the healthcare supply chain and create **robust logistics processes** that can mitigate the effects of health emergencies.

Overall, this study finds that the adoption of AI tools for improving MCMs production, stockpiling, distribution and supporting decision-making in these areas can enhance the EU's crisis preparedness and response capabilities by preventing MCMs shortages and ensuring adequate allocation of medical supplies. However, the study also highlights that the ability to integrate all the available data into AI models is still at an early stage in the EU and, until now, has been mainly restricted to electronic health records and medical images due to data availability and interoperability limitations. Addressing these constraints must be a priority for the deployment of AI tools within the EU that will lead to user acceptability and reliability.

In the context of the HERA Work Plan 2023, the results of this work will inform the actions of the HERA to develop, produce and distribute MCMs before and during a health crisis.

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List of abbreviations

Abbreviation	Description
MCMs	Medical countermeasures
AI	Artificial intelligence
EU	European Union
HERA	Health Emergency Preparedness and Response Authority
MS	Member States
EC	European Commission
WHO	World Health Organisation
ATHINA	Advanced Technology for Health Intelligence and action IT system
EMA	European Medicines Agency
RAG	Retrieval Augmented Generation
ML	Machine learning
IoT	Internet of Things
AVs	Autonomous vehicles
UAVs	Unmanned aerial vehicles
DL	Deep learning
GenAI	Generative AI
GMP	Good manufacturing practice
HPC	High performance computing
NLP	Natural language processing
LLMs	Large language models
GDP	Good distribution practice
GCP	Good clinical Practice
ECCPM	European Consortium for Continuous Pharmaceutical Manufacturing

Glossary

Term	Description
Artificial Intelligence system	Machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.
Automated Mobile Robots	An autonomous mobile robot is a type of robot that can understand and move through its environment independently by using set of sensors, AI, ML and compute for path planning to interpret and navigate through their environment, untethered from wired power.
Autonomous Vehicles	An autonomous vehicle, or a driverless vehicle, is one that is able to operate itself and perform necessary functions without any human intervention, through ability to sense its surroundings. An AV utilises a fully automated driving system in order to allow the vehicle to respond to external conditions that a human driver would manage.
Blockchain	The blockchain is a shared, immutable ledger that facilitates the process of recording transactions and tracking assets in a given network.
Deep learning	Subset of ML that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.
Digital twin	A digital twin is a virtual representation of an object or system throughout its life cycle, which is updated through real-time data and makes use of simulations, machine learning and reasoning to facilitate decision-making.
Generative AI	Generative AI is a type of AI that generates images, text, videos, and other media in response to inputted prompts.
Good Manufacturing Practice	Good Manufacturing Practice (GMP) is a system for ensuring that products are consistently produced and controlled according to quality standards. It is designed to minimise the risks involved in any pharmaceutical production that cannot be eliminated through testing the final product. GMP covers all aspects of production from the starting materials, premises, and equipment to the training and personal hygiene of staff.
High performance computing	Technology that uses clusters of powerful processors that work in parallel to process massive multi-dimensional data sets, also known as big data, and solve complex problems at extremely high speeds.
Internet of things	The Internet of Things describes the network of physical objects—“things”—that are embedded with sensors, software,

	and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.
Large Language Models	Large language models are unsupervised deep learning models of considerable volume, pre-trained on large amounts of data. The underlying transformer is a set of neural networks each consisting of an encoder and a decoder with self-attention capabilities. Encoders and decoders extract meanings from a sequence of text and understand the relationships between words and phrases in it and are able to process entire sequences in parallel.
Machine learning	Branch of AI and computer science that focuses on using data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.
Medical Countermeasures	MCMs include drugs, vaccines, biologics, sensors, diagnostics, personal protective equipment, wearables, and other technologies that can assist in the detection, diagnosis, prevention, or treatment of a chemical, biological, radiological, or nuclear threat.
Natural Language Processing	Natural language processing (NLP) is a branch of AI that enables computers to comprehend, generate, and manipulate human language. Natural language processing has the ability to interrogate the data with natural language text or voice.
Process optimisation	Process optimisation is the systematic approach of analysing and improving business processes to achieve maximum efficiency, effectiveness, and quality.
Retrieval Augmented Generation	Retrieval-Augmented Generation (RAG) is the process of optimising the output of a large language model, so it references an authoritative knowledge base outside of its training data sources before generating a response.
Smart contract	A smart contract is a decentralised computer program running on a blockchain network that automatically and deterministically executes agreements based on predefined conditions.
Stockpiling	The practice of storing a large supply of a good for future sale/use.
Supervised learning	Subcategory of ML and AI. It is defined by its use of labelled data sets to train algorithms that to classify data or predict outcomes accurately.
Supply chain	Supply chain is the network of all the individuals, organizations, resources, activities and technology involved in the creation and sale of a product. Supply chain encompasses sourcing raw materials, refining those materials into basic parts, combining those basic parts to create a product, order fulfilment/sales, product delivery, customer support and return services.

Unmanned Aerial Vehicles	An Unmanned Aerial Vehicle is a power-driven crew-less air vehicle. Usually UAV is guided autonomously, by remote control, or both, and incorporates sensors.
Unsupervised learning	Uses ML algorithms to analyse and cluster unlabelled data sets. These algorithms discover hidden patterns or data groupings without the need for human intervention.

1. Introduction

1.1. Study scope

The production and manufacturing processes of **medical countermeasures (MCMs)**, which include medicines, medical devices, and other medical equipment, have been evolving in recent years to become more sustainable through improved effectiveness and efficiency processes (European Commission, European Health and Digital Executive Agency, 2023). **Artificial Intelligence** (AI) can significantly contribute to this transformation by enhancing various aspects of MCMs production, distribution and stockpiling, as well as supporting policymakers and stakeholders in decision-making (L. Cao, 2023). In this respect, the EU4Health programme 2021-2027 is paving the way for a strong European Health Union, including among its key priorities reinforcing the European Union's resilience to cross-border health threats (DG SANTE, 2021a).

Following the onset of the COVID-19 pandemic, the European Union (EU) established in 2021 the **European Health Emergency Preparedness and Response** (HERA) authority aimed at contributing to the digital transformation of health systems. HERA plays an essential role in ensuring the **resilience of health systems** across the EU by promoting advanced research and development of MCMs and related technologies for tackling priority cross-border health threats. Since its creation, HERA has worked with Member States (MS) and various stakeholders, including industry and civil society, to enhance **preparedness and response to health threats**, particularly concerning MCMs. The procurement and stockpiling of MCMs constitute core components of HERA's mandate, involving acquiring critical medical items such as medicines and vaccines to maintain a high level of preparedness within the EU against future cross-border health outbreaks (DG SANTE, 2021b).

In this regard, the European Commission (EC) is developing a common EU **strategy for stockpiles and supplies** aiming to assess the existing stockpiling capacity at both the EU and national levels and ensure the effective geographical coverage and timely deployment of critical MCMs across the EU (DG SANTE, 2024). To support the development of this new strategy, it is essential to assess the use of AI-based tools and systems in production scale-up, stockpiling and MCMs, as well as in decision-making processes. This assessment will help identify how AI can be leveraged to enhance the resilience and responsiveness of MCMs supply chains, ensuring that critical medical supplies are available where and when they are needed most.

This report presents the results of a literature review and consultation activities focused on pinpointing the various areas within the **production scale-up, stockpiling and distribution of MCMs** where AI has been either implemented or holds the potential for **enhancing processes and logistics** in the context of **emergency preparedness and response**. Furthermore, it also examines AI-based tools and systems designed to **support decision-making** during health crises across the above-mentioned areas, spanning the European level down to regional and local contexts. Ultimately, this report provides an overview of the current availability of AI-based tools in MCMs production, stockpiling, distribution and decision-making and identifies the most significant bottlenecks and challenges hindering a swift response and its approach before a crisis hits.

1.2. Study objectives

This document aims to provide a scientific background and contribute to the development of a common EU strategy for stockpiles and supplies, to which HERA is contributing by assessing the stockpiling capacity of both the EU and MS and developing a strategy to ensure geographical coverage and timely deployment of critical MCMs across the EU (DG SANTE, 2024). Specifically, it will support HERA in identifying the **key areas and map the current use of AI-based tools across the MCMs production, stockpiling and distribution** where AI can be applied in the coming years to make EU preparedness and response to health emergencies more effective and efficient.

More specifically, this document has five main objectives:

1. Identify and map the different areas across the **production scale-up** of MCMs where **AI** is or could be applied to **improve processes** and **logistics** during a health crisis:
   - Improve manufacturing and **quality control processes** across the upstream supply chain for both pharmaceuticals and biopharmaceuticals.
   - **Production optimisation** and scale-up to **automate and optimise production** processes and allow for quick scale-up when needed in the management of health crises (e.g., massive vaccination campaign).
2. Identify and map the different areas where **AI** is or could be implemented to improve MCMs **stockpiling** and **distribution**:
   - **Inventory optimisation** and **automation of warehouses** to improve the allocation of MCMs and vaccines during health emergencies, prioritising vulnerable populations and preventing shortages.
   - Improved distribution and **optimal allocation** of MCMs during a health crisis and implementation of AI-powered applications to **deliver** MCMs in limited-capacity areas.
   - Anomalies detection through forecasting and detecting **panic buying** and pre-empting hoarding trends to adapt MCMs production, stockpiling and distribution of MCMs.
3. Identify and map AI-based tools that can **support decision-making** during health crises:

- Identifying and monitoring health threats and simulation of disease spread to implement and prioritise control measures.
- Guiding experts on MCMs demand, ensuring equal access to MCMs during health crises and deciding on the strategies to prevent MCMs shortages and accurate distribution and stockpiling.

4. Identify the most crucial **bottlenecks, barriers, advantages and disadvantages** to the **implementation** of **AI solutions** for MCMs production, stockpiling, distribution and decision-making in health emergency preparedness and response.
5. Describe the **availability** and the **implementation** of the identified AI tools.

1.3. Structure of the report

The following section of the report provides an overview of the state-of-the-art use of AI tools in MCMs production, stockpiling, distribution and decision-making in health emergency and response. This section provides the reader with an understanding of the potential uses of AI-based tools in the field and areas of potential implementation. Additionally, it reviews the main AI techniques and technologies aimed at improving MCMs production, stockpiling, distribution and decision-making.

Section 2 presents the results of the desk research, which includes the findings from both the literature review and consultation activities. Findings are organised into five main sections:

- AI for MCMs production scale-up in health emergency preparedness and response
- AI for MCMs stockpiling and distribution in health emergency preparedness and response
- AI supporting decision-making in health emergency preparedness and response
- Advantages and disadvantages of AI-based tools with a focus on improving MCMs production, stockpiling, distribution and decision-making for crisis preparedness and response
- Bottlenecks and barriers to the implementation of AI solutions to support crisis management and MCMs production, stockpiling and distribution in the EU

Section 3, “Discussion and conclusions”, is dedicated to the synthesis of the key findings and presents the major implications of the main results. This section also discusses the maturity and impact of the identified AI-based tools and suggests directions for future research.

Annex 1 includes the methodology followed in this report, detailing the key research questions, the literature search strategy, the data extraction methodology and the organisation of consultation activities.

1.4. Overview of the state-of-the-art

According to the World Health Organization (WHO), the field of health emergency and disaster preparedness refers to the knowledge and **capacities to effectively anticipate, respond and recover** from the impacts of likely, imminent or current hazardous events or conditions (PAHO, n.d.). In the European context, the WHO Regional Office for Europe has initiated the development of a new regional strategy and action plan to strengthen health emergency preparedness, response, and resilience in the WHO European Region, namely "Preparedness 2.0 (World Health Organization, 2024)". Among its objectives, this strategy aims to promote access to MCMs by ensuring that preparedness efforts focus on expedited research and development. This includes pre-negotiated benefit-sharing agreements, appropriate financing and regulatory instruments, scalable manufacturing platforms, and technology transfer agreements. Additionally, it seeks to establish emergency supply chains to enable **equitable access and support evidence-based decision-making** for preparing and responding to health emergencies. In parallel, the Advanced Technology for Health Intelligence and Action IT System (**ATHINA**), an IT system being developed by HERA, aims to collect data to detect potential health threats, offer early assessments and response options and monitor and address vulnerabilities in the supply of MCMs to ensure preparedness and effective response to public health emergencies (HERA, 2023).

The **appropriate distribution and provision of MCMs** during a health crisis is essential for coping with disruptions during health threats, and it depends on the existence of **resilient supply chains** (Hopp et al., 2022). AI can play a pivotal role in enhancing each stage of the entire supply chain management, given the instrumental role of data-driven processes and the potential of AI to enhance crisis management by offering swift, efficient, and reliable solutions and delivering intelligent and automated assistance (Van Wart & Kapucu, 2011).

In the aftermath of the COVID-19 global health crisis, there has been a concerted effort by international and national institutions to bolster **health crisis preparedness and response mechanisms**. The pandemic exposed significant **vulnerabilities** in health systems worldwide, prompting a re-evaluation of existing strategies and the adoption of more robust, tech-driven supply chains. Following the outbreak of COVID-19, the European Commission promptly took the lead in coordinating a unified European response. A central focus was **leveraging emerging technologies**, particularly AI-driven tools, to support the efficiency and effectiveness of health crisis management. For example, the EC and AI Alliance led a coordinated effort to embrace AI solutions, aiming to address public health and global crises after the outbreak of COVID-19. Initiatives like the Repository of AI and Robotics Solutions were launched to disseminate

knowledge on AI and accelerate the adoption of cutting-edge technologies to explore how they can assist in tackling specific tasks, including MCMs supply chain disruptions during a health emergency (European Commission, 2021; H. Wang et al., 2023).

The **MCMs supply chain** refers to the extensive infrastructure and processes involved in the manufacturing, production, distribution and stockpiling of MCMs. It encompasses the sourcing of raw materials, the manufacturing of vaccines, therapeutics, and medical devices, as well as their storage, transportation, and delivery to healthcare facilities or populations in need (Hopp et al., 2022). Organisations can adopt emerging technologies such as blockchain or AI to become more self-sustainable and resilient to cope with potential disruptions occurring in supply chains and mitigate the effects of health emergencies (L. Cao, 2023). Supply chain resilience can be defined as the capacity of supply chains to **cope with unexpected changes with minimum impacts on the overall process** and withstand crises, maintaining their continuity despite changes in the market (Hopp et al., 2022). In this sense, the principles of scale-down d~~ownscaling~~ can be applied to conduct assessments on mitigating the effects of health emergencies on the supply chain and can provide a better understanding of the required design space. Scale-down principles related to the MCMs supply chain involve the following strategies (Maramba et al., 2024):

- **Prioritisation**: Identifying essential healthcare goods and services to ensure critical needs are met first.
- **Efficiency**: Streamlining processes and logistics to optimise resource allocation and reduce waste.
- **Flexibility**: Building agility into the supply chain to adapt quickly to changing demands and circumstances.
- **Localising**: Increasing reliance on local suppliers and manufacturers to reduce dependencies on international supply chains.
- **Communication**: Enhancing transparency and communication among stakeholders to coordinate efforts effectively.
- **Resilience**: Strengthening the supply chain's ability to withstand shocks and disruptions through strategic planning and contingency measures.

The design space process involves **making strategic decisions** about different entities, systems and conditions that ensure quality for MCMs supply. According to the European Medicines Agency (EMA), the relationship between the process inputs (materials attributes and process parameters) and the critical quality attributes can be described in the design space. A design space can be defined in terms of ranges of material attributes and process parameters or through more complex mathematical relationships (EMA. Committee for Human Medicinal Products, 2017). Some considerations, such as what resources and facilitates are required to meet demands, assess market conditions, identify potential bottlenecks in the logistic process and develop contingency

plans for unforeseen events, have been proposed as a framework for making decisions about the structure of the supply chain (Mouchlis et al., 2021). AI has the potential to **support the design space process** by analysing large historical sale data sets, market trends and other variables to create real-time demand models, as well as help with plan production by considering factors such as customer changes, production capacities, resource availability and order priorities to minimise existing bottlenecks and optimise production efficiency. For example, high-performance computing (HPC) during the first phases of the COVID-19 pandemic was leveraged to pre-check medical measures against COVID-19. In that particular case, the strength was that the industry agreed to clear governance rules to share data, and this example could be used as a working precedent (Alizadeh-Meghrazi et al., 2022; Garcia & You, 2015).

Besides COVID-19, health crises significantly impact populations, necessitating urgent and coordinated resource mobilisation. These crises often reveal weaknesses in healthcare systems, such as slow processes, lack of supply chain visibility, non-standardised drug identification, inefficient stock management, poor drug traceability, and inadequate data utilisation (Ai et al., 2022; Chalmeta et al., 2023). Optimal management of a health crisis requires rapid and reliable feedback from the current extent of the threat, updated surveillance, and effective communication to respond effectively at each step (Bravo et al., 2023). AI-based tools can contribute by **expediting data analysis** and support policymakers and stakeholders in mapping out where the MCMs need to lie and how to proceed with distribution. However, AI-based tools, particularly in sensitive fields such as health and emergency response, introduce **ethical, societal, and technological challenges** that require careful consideration and assessment of widespread uptake within the EU. To ensure the widespread uptake of these tools within the EU, it is essential to address issues of fairness, explainability, and trust. Without addressing these pillars for real-life deployment, the feasibility of these tools and their alignment with EU values can be questioned, potentially undermining their inherent value for health crisis preparedness and response.

1.5. AI-based tools to improve MCMs production scale-up, stockpiling, distribution and decision-making

AI is a rapidly evolving tool revolutionising many aspects of healthcare. It is a vast field that includes a variety of methodologies, such as computer vision, natural language processing, and machine learning, and these methods are useful for pattern recognition, prediction, and large-scale data analysis.

In 2023, OECD member countries approved a revised version of the organisation's definition of an AI system (OECD, 2023). The updated definition reflects developments over the last five years, improves technical accuracy, and makes it more 'future-proof'. The EU's forthcoming AI Act has aligned the definition of AI to the updated OECD definition in its final text. In this study, we will

consider the updated OECD definition of 'AI systems' as a baseline to ensure technical accuracy and international alignment. The definition can be found in the box below.

***Box 1**. Working definition of "AI system"*

> "An AI system is a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment."

Moreover, for this study, several AI techniques and algorithms have been considered. Particularly, the OECD Framework for the Classification of AI Systems provides a comprehensive and structured approach that helps characterise AI systems based on key dimensions (e.g. Data & Input, AI Model, and Task & Output). The framework provided by the OECD has been used as a guide to establish a classification model by providing a systematic way to assess and categorise different AI systems based on their characteristics and the way they interact with one another. Ultimately, the framework has allowed the study team to identify the potential risks and benefits associated with different AI systems, as well as inform how these AI techniques can be used for further development of policies and regulations related to AI and MCMs production, stockpiling, distribution and supporting decision-making.

As a result, the study team considered four AI techniques. The table below provides an overview of the categories examined, highlighting their key characteristics and tasks. On one hand, symbolic AI models assist with inference and classification tasks, such as those found in knowledge-based and expert systems. On the other hand, statistical AI models enable trend and pattern prediction based on data, including methods like linear regression, logistic regression, decision trees, neural networks, naïve Bayes, and deep learning. The hybrid AI model integrates both approaches to extract complex features, enhance pattern recognition, and improve learning capabilities. Lastly, blockchain technology facilitates the simultaneous use and sharing of decentralised data.

***Table 1.** Definition of the AI techniques considered in this study.*

AI technique	Symbolic AI models	Statistical AI models	Hybrid AI models	Blockchain
Description	Symbolic or **knowledge-based** AI uses human-generated logical representations to infer a conclusion from a set of constraints (variables). These	**Statistical AI models** (e.g. genetic algorithms, neural networks and deep learning) identify patterns based on data rather than expert, human knowledge. They were previously	Many applied AI systems **combine symbolic and statistical** models into "hybrid" models. For example, NLP algorithms often combine statistical	Digital database containing information (such as records of financial transactions) that can be **simultaneously used and shared**

	constraints include **rules**, ontologies and search algorithms and rely on explicit descriptions of variables and descriptions of the interrelations between these variables.	used primarily for recognition, more recently, they are also being used for tasks like generation, such as synthesising and generating images or audio.	approaches building on large amounts of data and symbolic approaches that consider issues such as grammar rules.	within a large decentralised, publicly accessible network
Examples	Knowledge-based and expert systems	Linear regression, Logistical regression, Decision trees, Neural networks, Naïve Bayes, Deep learning	NLP	Authentication of MCMs, vaccines distribution framework
Data	Labelled and categorised	Unlabelled and uncategorised	Can be labelled or unlabelled	Unlabelled and uncategorised
Task	Making inferences and performing classification	Trend/ pattern prediction	Complex feature extraction and pattern recognition/ learning abilities	Manage transactions between entities securely and impartially/ data authentication/ smart contracts

Source: author's elaboration based on literature review findings.

Before 2019, AI has been predominantly employed in medicine and healthcare administration (Fontes et al., 2023; Nayarisseri et al., 2021). However, the COVID-19 pandemic shifted its focus from medicine to public health (Olawade, 2023). AI advancements are expected to play a key role in MCMs production and distribution in the future, which is an important component of pandemic preparedness (Chiu & Miller, 2019), helping to identify vulnerable areas within the supply chains, prevent MCMs shortages and improve response time, reducing the impact of future pandemics.

2. Results

In recent years, several AI-based tools have been developed to address these challenges. Even before the COVID-19 outbreak, the value of these technologies in preparing for and responding to health emergencies was recognised, with some applications already in use (Syrowatka et al., 2021; Vermiglio et al., 2022). However, the global pandemic marked a pivotal moment in the development of AI-based tools. These tools have since advanced in improving COVID-19 detection, vaccine process optimisation and distribution, predicting medical countermeasure shortages, and supporting decision-making for stakeholders (Gupta, 2023; Radanliev & De Roure, 2023a; Rosa et al., 2022).

This section summarises the main results of the literature review and the enrichments of these with the information gathered in consultation activities with key experts in the area. It starts with the set of identified AI tools for improving MCMs production, stockpiling, distribution and decision-making, while the last part is on the identified bottlenecks and barriers to the adoption of AI tools in the above-mentioned areas.

2.1. AI for MCMs production scale-up in health emergency preparedness and response

The main outcomes of the literature review on AI-based tools to improve MCMs production scale-up in health emergency preparedness and response are:

- ML techniques have been widely applied in **drug discovery and development** to expedite the analysis of vast amounts of available data and foster the **identification of new molecules** and compounds.
- The use of AI-based tools in cell line engineering offers potential benefits in developing genetically modified cell-lines and improving the **efficiency and scalability of processes** to advance personalised medicine and tailored therapies as well as supporting protein sequence optimisation..
- AI-based solutions are used during health emergencies to effectively **automate the production processes** of MCMs, reducing manufacturing time and cost. In this respect, blockchain can be leveraged to ensure the **quality and safety of products.**

MCMs **production scale-up** refers to the process of increasing the manufacturing capacity of MCMs to meet higher demand levels from pre-clinical phases to commercial production. When new MCMs are developed, they typically undergo initial production at a very small scale for testing and validation purposes and increasing scales to supply clinical demand. Production scale-up

involves various steps, including: **1) drug development, 2) sourcing raw materials, 3) optimising manufacturing processes, 4) increasing production capacity,** and **5) ensuring quality control measures are maintained throughout the expanded manufacturing process** (Gama et al., 2022). However, as demand increases, particularly during public health emergencies or outbreaks, production must be increased to ensure an adequate supply is available to meet the production cadence. Thus, production is typically conducted at a defined scale to accommodate commercial processes. Once the maximum batch size is reached, the frequency of batch manufacturing is adjusted to meet demand.

AI-based tools have shown their potential to improve specific key areas of production scale-up and manufacturing processes, such as drug development and design and improved automation in MCMs manufacturing and quality control processes (Gupta, 2023). In the context of health emergency preparedness and response, most of the studies identified within the literature focus on exploring the role of AI in developing new treatment options during the COVID-19 pandemic, and little has been published concerning how AI could support cell-line production, protein sequence optimisation and small molecule synthesis optimisation (Kolluri et al., 2022). In this regard, this literature review allows us to explore its implementation in less-targeted areas.

In the era of big data, machine learning (ML) and deep learning (DL) approaches have been used as an efficient way to **deal with huge amounts of generated data in drug development,** including processes such as compound screening, drug design, biomarkers identification and selection and identification of novel drug targets for various diseases (Paul et al., 2021). According to the literature, different AI technologies, mainly ML techniques, can be applied to mine and analyse large datasets and multi-omics building from existing validated data to provide information on the potential structure and function of biological targets and predict their role in a disease pathway. While early target identification and prioritisation is a critical step where ML could help **improve the efficiency and effectiveness of drug development**, the evidence suggests it is important to validate the role of the biological target in the disease of interest through subsequent studies using other AI techniques or conventional methodologies. ML and DL have also been used to predict cancer cell sensitivity to chemotherapy, track cell movements, and predict interactions between compounds and cell lines, thus improving efficiency and precision in drug discovery, learnings that could potentially be applied in the field of emergency response (Geethu S & Vimina ER, 2023).

Different ML techniques, such as generative adversarial networks and computer-aided synthesis prediction models, have been used to **design novel molecules and enhance small molecule synthesis optimisation with desired therapeutic properties to optimise and develop new treatments and vaccines** (Huang et al., 2023; Johansson et al., 2019; Radanliev & De Roure, 2023a). To improve existing treatments, ML models have shown efficacy in identifying structural

or chemical patterns that correlate with specific compound properties for creating compound library screening and scoring the ease of synthesis of compounds (Han et al., 2023; Stanley & Segler, 2023). Finally, AI approaches are effective in elucidating drug-target interactions and provide predictions about the interaction between previously unknown drugs or identifying mechanisms of action, which may help predict the toxicity of a molecule based on specific known features. According to the literature, this application of AI can help guide drug repurposing efforts based on previously characterised compounds (Niazi, 2023). In addition, AI tools have been used to improve drug toxicity prediction as they provide more accurate and efficient methods for identifying the potentially toxic effects of new compounds before they are tested in human clinical trials, thus saving time and money (Tran et al., 2023; Vijayan et al., 2022).

ML techniques such as DL and neural networks have also been applied in pharmaceutical manufacturing processes by **leveraging automation processes,** which is crucial during health emergencies. The implementation of ML techniques has allowed advancement in cell-line development, emphasising the **acceleration of processes and reduction of time** (Nayarisseri et al., 2021). For example, the utilisation of automated systems plays a pivotal role in supporting meeting regulatory standards (Shaw et al., 2018) and allowing the analysis of simultaneous cell lines, boosting the efficiency of cell culture analysis and allowing the integration of novel research areas such as the implementation of nanofluidic systems (Shaw et al., 2018). Moreover, AI has been reported to be used to shorten the timeline for cell line development and, therefore, **decrease the resources invested in the process,** which can be useful during a health crisis (Cavasotto & Di Filippo, 2021).

Evidence has also shown that ML models using process development data allow the **identification of optimal processing parameters,** thus facilitating **quicker process scaling** while **minimising waste and development time**. Digital twins, which are virtual replicas of real-world processes, have been employed for process design optimisation to evaluate, forecast, and enhance process performance. Digital twins have shown to be particularly useful in analysing manufacturing processes with limited development data (Huang et al., 2024; Niazi, 2023). Furthermore, digital twins, in combination with ML techniques such as neural networks, have been deemed useful to **implement advanced process control** (Huang et al., 2023). Evidence has also shown that these methods can be applied to develop control processes to predict whether a process is performing under a state of control by using ML tools, in combination with real-time sensor data, to **improve existing manufacturing line efficiency**. These models leverage the data needed for development while enhancing robustness; however, they have so far been used only in academic contexts." (C. Wang et al., 2023).

Besides ML, **blockchain** has also shown its potential to enhance MCMs production scale-up by ensuring **quality, authenticity and safety**. An example is the blockchain-based platform that provides a solution to control the quality of MCMs through product specifications or supply

volumes (Hu et al., 2023b). This platform combines blockchain, the Internet of Things (IoT), and ML, allowing real-time monitoring of vaccine quality in terms of effectivity and activity. In addition, blockchain technology has been shown to **support the identification of vaccine counterfeiting** and ensure the provenance, handling and safety of vaccines. To do so, Sarfraz et al. (2022) conducted a sentiment analysis using ML on vaccine reviews, enabling accurate interpretation of the vaccine review data stored in the blockchain platform and facilitating enterprises to **improve vaccine quality** and address consumers' concerns (Sarfraz et al., 2022a). According to the authors, the continuous training of the ML model will improve the accuracy of the sentiment analysis to provide a longitudinal quality control process of vaccines. This model provided a high accuracy rate of nearly 80%, providing strong support for enterprise credibility evaluation and consumer decision-making (Hu et al., 2023b). AI-driven tools have also been used in combination with other cutting-edge manufacturing technologies to achieve better results by implementing **Industry 4.0**. Industry 4.0. describes the fourth industrial revolution, which combines rapidly evolving technology and may provide a well-controlled, hyper-connected, and digitalised pharmaceutical value chain for the producer (Niazi, 2023).

An application of Industry 4.0 using blockchain technology is the **Vaccine 4.0 framework** (Sarfraz et al., 2022a). Studies focused on COVID-19 vaccines suggested that it could allow for improved production scale-up of vaccines by **estimating real-time demands** using AI and ML applications and setting up a blockchain system. However, the literature suggested that to be expanded, this framework requires the digitalisation of manufacturing, production, and distribution processes which is not currently done in many countries. The benefits that digitalisation can bring include continuous assurance of process control and potentially improving the quality of large-scale vaccine production, cost-effectiveness, and manufacturing processes (Sarfraz et al., 2022a). While these models offer improved robustness and leverage necessary data for development, concerns remain regarding data security. The increased flow of sensitive data to a blockchain could introduce heightened risks of hacking and unauthorized access.

Besides the potential role of AI-based tools in enhancing drug discovery and development, the evidence highlights the need to give careful attention to **ethical and regulatory considerations** such as data privacy, consent, and the responsible handling of patient data. Guaranteeing the **anonymisation of patient information** utilised in the development phase of AI-based tools and ensuring that proper consent has been obtained for the utilisation of their data in research are key aspects to be considered for the deployment of AI-based tools in the EU (Rosa et al., 2022).

Although several areas of the MCMs production scale-up have been improved by the implementation of AI-based tools there are still potential areas that could benefit from further AI development in health emergency preparedness and response. According to the literature, using AI and ML in cell line engineering could significantly impact the development of genetically modified cell lines for various purposes, such as studying virus replication (Sarfraz et al., 2022a),or

increasing the understanding of manufacturing gene therapies. However, the state-of-the-art methods for many of these processes are currently based on a very limited mechanistic understanding. Additionally, automating manual steps involved in preparing, reprogramming, and verifying induced stem cells can enhance efficiency, scalability, and reproducibility. This automation can be particularly beneficial for addressing health threats like infections or viruses that require a high consumption of cells (German Cancer Research Center, 2023). Furthermore, according to the existing evidence the integration of AI-based tools has the potential to advance personalised medicine and targeted therapies by speeding up progress and enhancing decision-making in fields such as molecular and cell biology, pharmacology, and toxicology (Deng et al., 2022; Hu et al., 2023b). The advancement of AI-powered applications in personalised and predictive health should always follow ethical compliance, as these applications give rise to ethical issues especially related to data safety.

Box 2. *Considerations from the experts: AI for MCMs production scale-up in health emergencies*

Highlights resulting from Experts' consultation

Experts from both academia and industry agreed on the importance **of implementing the use of digital twins for process design optimisation**. Experts explained that digital twins can be leveraged to test different scenarios and decide which is the best option. In the context of product design optimisation (vaccines and drugs), digital twins are considered beneficial for analysing manufacturing processes characterised by a limited amount of development data, where AI/ML models could potentially leverage prior knowledge of the product and process (e.g., from previous studies, development programs, and scientific literature) to more quickly **identify the optimal processing parameters,** thus **reducing design time and waste**.

One expert from the industry mentioned that digital twins can also be used for MCMs evaluation and clinical studies to **predict the efficacy and safety of pharmaceuticals**. Furthermore, he explained that they can aid in the development of clinical studies aimed at identifying characteristics of specific patient groups. This can be achieved by **simulating various scenarios** using digital twins and subsequently comparing the outcomes of each scenario, as well as identifying groups that may be at higher risk.

An expert with academic and pharmaceutical industry background provided a detailed description of how AI-powered tools can enhance MCMs production and quality management. Pharmaceutical companies must adhere the Good Manufacturing Practice (GMP) standard and **adhere to quality management processes** and ensure that appropriate controls are in place. The expert explained that AI-driven tools can help integrating large quantities of process data and release early warnings, if necessary, in terms of problems or issues in the MCMs production process such as lack of raw materials or production interruption. This last consideration including the use of large datasets was also mentioned by several experts regardless of their backgrounds. Experts also mentioned that AI could also be used to model where bottlenecks in processes occur under stress, what mitigation could help and what disaster recovery measures could be performed.

Lastly, **experts from the industry highlighted that** AI-based applications can support drug development as long as **data and evidence are available**. For example, for the improvement of current therapies or new treatments, there must be some evidence available on the disease, signs and symptoms to be tackled. Further, AI can help improving software pipelines for protein structures forecasting, which is a very

promising field of application in current years. AI-based applications can also be leveraged for post market surveillance.

The table below provides an overview of the main AI-based tools identified for MCMs production scale-up in health emergency preparedness and response, the AI technique utilised, and the level of maturity and implementation in the market (Table 2).

Table 2. *Main areas of AI implementation in MCMs production scale-up, enabling technologies and AI-based tools were identified through the literature review*

Area of implementation	Enabling technologies	Functioning and goals	Availability[a]	AI-based tool author
Drug discovery/ development	Deep-learning frameworks linked to distinct molecular representations	Deep-learning model to forecast the properties of a molecule based on its structure	In development	(Huang et al., 2023)
Small molecule synthesis optimisation	Recurrent neural networks	SMILE: recurrent neural network for in de novo molecular design. It is used to analyse and model molecular sequence data. SMILE can generate new sequences that follow the conditional probability distributions learned from the trained model	On the market	(Arús-Pous et al., 2019)
	Deep ResNet architecture for predicting inter-residue distance and dihedral angles	Protein structure prediction model to optimise compound design to modulate desired interactions	Research use	(Geethu S & Vimina ER, 2023)
	Virtual screening and de novo design	Generative model to create compound virtual library screening and score the ease of synthesis compounds	Research use	(Stanley & Segler, 2023)
	Computer-aided synthesis planning	Prediction model to find routes to synthesise novel compounds and optimise routes for existing chemical compounds	Research use	(Johansson et al., 2019)
Support MCMs manufacturing processes and quality control	Industry 4.0: Blockchain, ML, IoT	COVID-19 Vaccine 4.0 framework: this network help in predicting failures, enable maintenance processes autonomously, self-organise logistics, and react to changes in production	Research use	(Sarfraz et al., 2022a)
		Framework to support the MCMs quality control (product specifications, supply volumes)	Open source	(Hu et al., 2023b)
	Bayesian optimisation and gaussian processes	Bayesian optimisation method to improve the mRNA IVT reaction and reduce production time and costs of COVID-19 vaccines	Research use	(Rosa et al., 2022)
	AutoAI algorithm	Self-adaptative forecasting of MCMs production and supply chain bottlenecks during pandemics	Research use	(Radanliev P et al., 2022)

[a] *For this study the availability of the tool in the market was defined as: "Research use" for AI-based tools developed within the academia but not yet implemented or tested within a real-world setting, "In development" for AI-based tools currently under design or improvement. Overall, the methodology of these tools has been published in the literature because of their innovative approaches, "Open source" for AI-based tools for which the original source code is made freely available and can be redistributed and modified, and "On the market" for AI-based tools that have been already deployed in real-world settings and are currently being used.*

2.2. AI for MCMs stockpiling and distribution in health emergency preparedness and response

The following sections describe the areas in which AI can contribute to improving MCMs stockpiling, distribution, and detection of sales anomalies and **Table 3** summarises the identified AI-driven tools in these areas.

2.2.1. AI for MCMs stockpiling

The main outcomes of the literature review on AI-based tools to improve MCMs stockpiling in health emergency preparedness and response are:

- ML-powered solutions are employed **to forecast vaccine and MCMs demand, improving stock management and resource** allocation while **preventing shortages** or overproduction **and waste management**, with a focus on optimising daily vaccination planning by means of predictive analytics and modelling techniques.
- AI and blockchain-based tools **enhance the management of MCM orders and inventories** during health emergencies, providing fast, secure transactions and improved traceability

Overall, the evidence agrees that integrating AI technologies within inventory management systems is key for maintaining an efficient MCMs supply. The literature shows how these inventory management systems can benefit from AI-powered techniques to forecast **MCMs future demands** and **optimise ordering** timelines and purchases based on supply chain dynamics and decentralised systems (Bandhu et al., 2023).

To prepare and have a more resilient healthcare system during an emergency, one of the preferred measures is the MCMs stockpile. **Stockpiling facilitates prompt responses to public health threats** to bolster healthcare system preparedness and is used to help **prevent shortages** and **ensure accessibility** to treatments and protection measures (Radanliev & De Roure, 2023a). To this end, AI-based tools based on a wide range of methods and techniques such as ML, blockchain and smart contracts have been developed to optimise the warehouse stockpiling of MCMs (Kamran et al., 2023) and improve inventory management processes (W. Cao et al., 2022). One of the applications of AI in stockpiling has been the development of ML algorithms to **analyse historical consumption** patterns and real-time data to adjust stockpile compositions and distribution strategies, ensuring prompt response when facing a potential health emergency (McDaniel et al., 2023). For instance, a supervised ML algorithm was developed to predict drug shortages and manage stockpiling in pharmacies and hospitals, allowing pharmacists to optimise their orders and inventories (Adulyasak et al., 2023; Gomasta et al., 2023; Pall et al., 2023). Besides

ML, evidence shows that the combination of AI-powered and IoT solutions can also improve MCMs stockpiling management systems by **identifying vulnerabilities** within the stockpiling and allocation systems and can **suggest risk mitigation strategies** based on real-time data monitoring. These tools are useful for stakeholders in **identifying areas for improvement** as well as **suggesting measures to prevent shortages or stock-outs** (Goodarzian et al., 2023).

Several ML algorithms aimed at **forecasting vaccine and MCMs demands** were developed during and after the COVID-19 pandemic to better estimate MCMs demands and prevent future shortages in the event of a health emergency (Arora et al., 2021; Kamran et al., 2023; Nguyen et al., 2021). According to the evidence, most of these ML algorithms use **predictive analytics** and modelling techniques to estimate the demand for MCMs based on **epidemiological data, population demographics, and historical usage patterns**, to allow better management of stock and **allocation of resources** and prevent shortages or **overproduction**. DL techniques have been employed for MCMs localisation to update in real-time the stocks in cold storage warehouses, which reinforces the traceability and quality assurance of products (Wu et al., 2023). One of the most complex models identified in the literature review aimed at improving the planning of daily vaccinations by generating decisions regarding quantities transferred between locations, inventory profiles of central nodes, vaccination centres and daily vaccinations. This model included disease-specific features, such as cold storage technologies, limited shelf life of mRNA vaccines under refrigerated conditions and demanding vaccination targets under extreme time pressure, to increase the efficiency and generalisability of the model (Georgiadis & Georgiadis, 2021).

Besides ML, blockchain solutions have also been applied to improve different areas of MCMs allocation and stockpiling such as **security, traceability, and accountability.** However, most of the AI-based using blockchain are focused on vaccines. For instance, a vaccination record platform was developed where the blockchain system saves the vaccination record and makes it available for healthcare professionals to improve healthcare delivery (Gomasta et al., 2023). According to the authors, this system ensures vaccination traceability through the **mutual authentication system** and **guarantees data integrity** (Liao et al., 2015). Blockchain has also contributed to improving forecasting MCMs demand since data transparency in blockchain systems provides **traceable transmission of information avoiding the risk of the "bullwhip effect"** and allowing estimating of real-time demands (Fontes et al., 2023; Shah et al., 2019). Another example identified during the literature review is the use of a blockchain network for **counterfeit detection of COVID-19 vaccines** (Munasinghe & Halgamuge, 2023). This network identifies the number of stocks that have arrived in the system and the exact location of the stock. The network showed similar results to existing supply frameworks, allowing distribution companies to manage their supply operations efficiently and securely, using an authorised distribution network (Munasinghe & Halgamuge, 2023).

Moreover, blockchain technology has also been used to develop **traceable vaccine management systems in combination with smart contracts**. According to the authors, both techniques facilitate **fast and secure transactions** and improved traceability to prevent shortages and enhance the management of MCMs orders and inventories. In addition, it has been observed that smart contracts **foster agreement execution** by allowing all stakeholders involved to be updated instantly, without an intermediary or consuming procedures (Bandhu et al., 2023). To do so, it has been proposed to use the EdDSA algorithm for digital signatures and the Burrows-Abadi-Needham logic to determine whether the exchanged information is trustworthy and secured when using smart contracts including MCMs (Ai et al., 2022b). However, as mentioned before, most of the identified AI-based tools were focused on vaccines suggesting the need to expand research to the implementation of these technologies in other MCMs (Gomasta et al., 2023; Pall et al., 2023).

Interestingly, some of the identified AI-based tools combined blockchain, IoT and ML to improve specifically the efficiency of the COVID-19 vaccine supply chain management system which according to the authors enabled decision-making and supported stakeholders (Gao et al., 2023; Hu et al., 2023a). For example, the transparency of blockchain technology fosters trust between stakeholders, real-time tracking of vaccine status via IoT monitors vaccine quality, and finally, ML models can predict vaccine demand and perform sentiment analysis on vaccine reviews to help companies improve vaccine quality (Hu et al., 2023a). Other AI technologies have also been applied to optimise the vaccine supply chain networks such as the Modified Whale Optimisation Algorithm with Variable Neighbourhood Search model, which was developed for COVID-19 vaccine allocation, inventory control, and distribution, has demonstrated superior performance in simulations compared to other algorithms. This superiority is based on parameters such as disease duration, quarantine length, average death rate, transmission rate, and lockdown level (Kamran et al., 2023).

Finally, the available evidence agrees that **AI-based tools support the sustainability of MCMs distribution and optimise vaccination stockpiling**. One form of supporting sustainability is by developing vaccine waste reverse supply models to address vaccine waste through optimisation of vaccination stockpiling, order processing and distribution which are responsible for a large portion of total costs (Amani Bani et al., 2022; Shadkam, 2022). The **incorrect estimation of vaccine demand is one of the major causes of vaccine expiration and increased costs**. In the context of the COVID-19 pandemic, vaccination demands were uncertain and accurate predictions using blockchain and ML algorithms could have resulted in considerable vaccine supply cost savings (Hu H et al 2023).

2.2.2. AI for MCMs distribution

The main outcomes of the literature review on AI-based tools that can support with MCMs distribution in health emergency preparedness and response are:

- ML models can contribute to the **optimisation of MCMs allocation processes** according to real-time demand, ensuring the availability of therapies, vaccines and other medical products at the right time and amount at the distribution points.
- AI-based solutions **help mitigate MCMs shortages** in healthcare hubs ensuring efficient allocation of resources and timely delivery of vaccines and essential MCMs-**based on real-time demand estimations**. Blockchain can complement this type of solution by ensuring safe and traceable resource transfer.
- Routing algorithms, autonomous vehicles (AVs) and unmanned aerial vehicles (UAVs) powered by AI techniques offer a new paradigm within the MCMs distribution and can be used to **relieve world load to healthcare task forces** during health crises as well as to reach remote areas or regions with limited capabilities.

By leveraging the capabilities of AI, different techniques have been applied to **improve MCMs distribution** and **allocation strategies** during health emergencies (Georgiadis & Georgiadis, 2021).

Evidence has shown that ML models can contribute to enhancing the automation of MCMs distribution processes and warehouse inventories. Predictive algorithms such as mixed integer logistics models were proposed to optimise the COVID-19 vaccine supply distribution (Goodarzian et al., 2023; Radanliev & De Roure, 2023a; Shahparvari et al., 2022) and the distribution of COVID-19 therapies like plasma therapies (Shirazi et al., 2021). According to the literature, predictive models enable the estimation of real-time and changing demands for MCMs, automating distribution processes in warehouses and logistics networks, and guiding them towards the final administration points. Moreover, predictive models play a crucial role in **allocating healthcare resources**, such as hospital beds, ventilators, and medical personnel. This can be achieved by leveraging real-time demand forecasts, responding to medical emergencies, and respecting capacity limits. Additionally, non-linear mathematical models have also been developed to determine optimal vaccine allocation strategies, minimising wastage and **promoting equitable distribution across populations** (Enayati & Özaltın, 2020).

Blockchain-based solutions have also been developed for **optimising drug distribution mechanisms in hospitals** by forecasting MCMs demand. In these cases, demand is forecasted by analysing historical data, demographic information and socioeconomic factors and by predicting fluctuations in demand, healthcare organisations and government agencies can adjust their distribution strategies during health emergencies (Bandhu et al., 2023; Huang et al., 2023; Liao et al., 2015). Other applications of blockchain in MCMs distribution are aimed at **improving** Click or tap here to enter text. **and supporting vaccine distribution in limited-capacity regions**

to facilitate targeted distribution efforts in affected areas. By prioritising high-risk areas and vulnerable populations, blockchain systems can ensure that MCMs are distributed equitably and effectively to those who need them most. In striving for equitable access to vaccines, the **Vaccine 4.0 blockchain system** has been reported to address anticipated obstacles in manufacturing and impartially distributing COVID-19 vaccines. Additionally, it employs **biometrics** to identify unregistered individuals who have become lost in the healthcare system, furnishing them with digital identities or twins. This approach was useful to guarantee that digitally undocumented populations received the COVID-19 vaccine, thus contributing to the broader goal of achieving widespread vaccination coverage overall (Sarfraz et al., 2022b). On the other hand, according to the literature, these applications using **biometric data might raise serious concerns for vulnerable populations** such as unregistered migrants and citizens' privacy that might be considered and prioritised in the updated versions of these tools.

Another observed use of tools combining blockchain and ML are the ones identifying optimised routing to facilitate faster distribution of MCMs during a health emergency. It has been shown that these AI-powered algorithms can play a crucial role in optimising transportation routes and logistics to ensure the timely delivery of MCMs to healthcare facilities, pharmacies, and distribution centres. These algorithms consider various factors such as geographical location, transportation infrastructure, and demand forecasting to minimise response times (McDaniel et al., 2023; Twaróg et al., 2021; Zabinsky et al., 2021). Although AVs, UAVs and other robotic applications are not yet widely adopted in healthcare settings, they are increasingly being considered for MCMs delivery, particularly in areas difficult to access or where traditional transportation methods may be limited. For instance, drones have the potential to fly over areas heavily affected by emergencies or disasters and automated mobile robots can be suitable to perform tasks that **facilitate the distribution of medicines** to patients in hospitals and collect patient data allowing hospital staff to dedicate more time to direct patient care as well as safeguarding professionals from exposure and preventing the spread of infections. (Deo & Anjankar, 2023). Furthermore, robots and AI-powered sensors used to secure thermostability can support the distribution of vaccines in regions with limited capacity (Deo & Anjankar, 2023; Sarfraz et al., 2022b).

2.2.3. AI for detecting sales anomalies

The main outcomes of the literature review on AI-based tools to detect MCMs sales anomalies in health emergency preparedness and response:

- AI-driven tools, mainly based on ML, can contribute to the early detection of sales anomalies, **panic buying and pre-empting hoarding trends**.

- Sales data can be integrated into the supply chain to **inform on real-time needs**, forecast stockpiling and future demand

According to the literature, in the aftermath of the COVID-19 pandemic, AI-based tools were developed to **detect sales anomalies**, such as panic buying and pre-empting hoarding trends, to help retailers **better estimate product demands and prevent MCMs shortages by detecting sudden spikes in demand or unusual purchasing patterns** (Adulyasak et al., 2023). For instance, an unsupervised ML model was built to identify anomalous sales observations within the first months of the COVID-19 pandemic. The model was used to inform distributors of real-time needs, forecast stockpiling and future demands, and **automate purchases predicting panic buying** (Georgiadis & Georgiadis, 2021). In addition, other publications propose to integrate sales data within the supply chain to provide a more comprehensive understanding of demand flow. By merging sales anomalies with inventory data, **AI-based tools can anticipate and respond to sudden fluctuations in MCMs sales** (Hu et al., 2023c). Evidence shows that ML models can be trained to recognise patterns of panic buying for specific products, geographic regions, or population groups. According to the authors, by continuously learning from data and refining their algorithm, these models can become increasingly accurate at detecting anomalies over time and could potentially be applied to other health threats (Adulyasak et al., 2023; Syrowatka et al., 2021).

In the past years, HERA has contributed to the development of the whole range of MCMs for pandemic preparedness, including communication with wholesalers to act if sales anomalies within the EU market are detected (Directorate-General for Health and Food Safety, 2022). In addition, the EU has also adopted regulations aimed at tackling future public health emergencies such as the Council Regulation (EU) 2022/2372. This regulation establishes a **framework of measures for ensuring the supply of specific MCMs in the event of a public health emergency at the EU level.** The Regulation includes the creation of the Health Crisis Board to coordinate the implementation of measures adopted under the Regulation. In the event of a public health emergency, Regulation 2022/2372 grants power to the Council to adopt, upon proposal of the Commission, a regulation activating the emergency framework and setting out appropriate measures to handle the situation. The main measures considered are:

- Inventory of MCMs production and production facilities: the EC may request manufacturers of crisis-relevant MCMs, to inform on their production capacity and existing stocks, in the EU or third countries. The information may also include a schedule of production plans for the next three months.
- Measures to ensure availability and supply: where is considered that there is a risk of shortage for MCMs, the EC may adopt measures to increase their availability such as reorganising supply chains and utilising existing stocks.

Box 3. *Considerations from the experts: AI for MCMs stockpiling and distribution in health emergencies*

Highlights resulting from Experts' consultation

Most of the consulted experts agreed that time series forecasting is the preferred technique to predict shortages. Experts from governmental authorities described that time series forecasting could be very useful for predicting disease outbreaks and shortages in medical supplies and overcome them in a structured way, by planning production and provision based on real-time-adjusted forecasts. According to the experts, this kind of applications were tested during the COVID-19 health emergency and have proven to be accurate.

Experts from the industry mentioned that AI-based tools can be useful to identify buying trends and **anticipate demands**. In addition, experts from governmental sector added that **using customs data and other public sector databases from different MS** could be leveraged by these tools to **anticipate buying trends and adjust production** and stockpiling of MCMs during health emergencies. However, they are aware that data availability and interoperability limitations could pose a strong barrier to this AI utility.

Surveillance systems and predictive AI solutions were deemed of utmost importance by governmental experts. According to them, these tools can automatically get and analyse population data and identify differences among population groups and companies purchases, compare them to the real stock to identify potential shortages and activate early warnings to key stakeholders and producers on what is expected to happen.

Lastly, most experts from all backgrounds agreed on the need to strengthen collaboration between stakeholders. In addition, three experts from governmental sector proposed that a **common European MCMs distribution system would be extremely useful** since some of the main barriers and challenges are given by the different MCMs distribution procedures and guidelines followed in the different countries. Having **standardised procedures for certification of compliance of MCM,** agreeing on a common procedure to collect, process and organise data about MCMs, selecting information to be disclosed from every European healthcare system and building a common MCMs distribution system could solve some of the key challenges in both AI research and health emergency management.

The table below provides a quick overview of the main AI-based tools identified for MCMs stockpiling and distribution in health emergency preparedness and response, specifying the AI technique used, and the level of maturity and implementation in the market (Table 3).

***Table 3**. Main areas of AI implementation in MCMs stockpiling and distribution, enabling technologies and AI-based tools identified through the literature review.*

Area of implementation	Enabling technologies	Functioning and goals	Availability[a]	Authors
Prevent drug shortages of MCMs and detect panic-buying	Blockchain, unsupervised models	Platform to track the demand and supply of MCMs and identify anomalies in sales transactions such as panic buying	Research use	(Adulyasak et al., 2023)
	Decentralised framework	PharmaChain: decentralised hyperledger fabric framework that enables on-chain and off-chain storage with smart contracts to improve drug traceability across the pharmaceutical supply chain	Research use	(Gomasta et al., 2023)
	eXtreme Gradient Boosted trees	Model to predict shortages for most drugs in the most-dispenses groups. It allows pharmacists to optimise their orders and inventories and reduce the impact of drug shortages	Research use	(Pall et al., 2023)
Improve the stockpiling and distribution of MCMs	Blockchain	Blockchain powered system aimed at improving the traceability of vaccine distributions and supporting vaccine allocation in limited capacity regions	Research use	(W. Cao et al., 2022)
		Ethereum blockchain-based solution for MCMs tracking and distribution based on smart contracts. The system allows a flow of MCMs via blockchain without intermediaries to forecast MCMs demands and optimise ordering timelines	Research use	(Bandhu et al., 2023)
		Traceable vaccine management system using smart contracts to store the records generated during the whole process from vaccine production to vaccination	Research use	(Ai et al., 2022c)
		Integer linear programming model to generate optimal decisions regarding the MCMs quantities allocated between locations, the inventory profiles of central hubs and vaccination centres and the daily vaccination plans	Research use	(Georgiadis & Georgiadis, 2021)
	Blockchain, ML, IoT	System combining blockchain, IoT, and ML (RNN, LSTM, and GRU models) that effectively address the real-time demands of COVID-19 vaccines and ensure vaccine quality and transparency among stakeholders.	Research use	(Hu et al., 2023c)
		System powered with the help of mobile and desktop applications, to monitor remote temperature and humidity, detect anomalies in cold	Research use	(Wu et al., 2023)

		warehouses enable customer interaction.		
	The computation technology used included a simulation, a neural network (NN), and a genetic algorithm (GA)	Simulation algorithm including a neural network and a genetic algorithm to create a hospital supply distribution chain to improve ordering strategies to deal with drug shortages	Research use	(Liao et al., 2015)
	Regression modelling	Earth Cognitive System for Coronavirus Disease 2019 is based on a geographic information system designed to support healthcare institutions in stockpiling medical equipment logistic needs and forecasting regional medical supplies	Research use	(Atek et al., 2023);
	Cuckoo optimisation algorithm	Network based on reverse logistics to improve manage vaccine wastes, minimise costs associated to wastes and optimise stockpiling	Research use	(Shadkam, 2022)
	Lagrangian relaxation algorithm	Model to design a COVID-19 vaccine waste reverse supply chain, minimise the system's total cost and carbon emission	Research use	(Amani Bani et al., 2022)
Optimisation of distribution routing	Receiving-staging-storing-distributing (RSSD) algorithm	Receiving-staging-storing-distributing metaheuristic algorithm to identify time and capacity compliant delivery routes for the distribution of MCMs to points of distribution during a bio-emergency	Research use	(McDaniel et al., 2023)
	Route Optimization Tool (RoOT), Vehicle Routing and Scheduling Algorithm (VeRSA)	Tool to optimise and plan routes for MCMs distribution accounting for transit time including temperature-sensitive products such as vaccines. This algorithm includes multi-day routing and integration with a mapping software to facilitate distance calculations and visualisation of routes.	Open source	(Zabinsky et al., 2021)
	Multiple probabilistic traveling salesman problem algorithm	Algorithm to adapt routing algorithms on the changing demand of costumers.	Research use	(Twaróg et al., 2021)
Equity in COVID-19 vaccine distribution	AI-powered sensors and robotics	Biosensor to support thermostable COVID-19 vaccine shipment and distribution in limited capacity regions. These sensors can detect and communicate when vaccines are defiled	In development	(Sarfraz et al., 2022b)
	Automated mobile robots	Robots to address the shortage of staff, keep a trail of records and place orders on time. They ensure that availability of medicines when needed	On the market	(Deo & Anjankar, 2023)

	Blockchain	Vacledger framework for supply chain traceability and counterfeit detection of COVID-19 vaccines. The framework handles the rules and regulations of vaccine importing countries and provide authorisation for cross the borders, register new and imported vaccines and find the number of stocks that have arrived in the Vacledger system	Research use	(Munasinghe & Halgamuge, 2023)
Automatization of warehouses and distribution centres	IoT, ML	Responsive-green-cold vaccine supply chain network to enhance the accuracy, speed, and justice of COVID-19 vaccine injection.	Research use	(Goodarzian et al., 2023)
	AutoAI algorithm	The *algorithm* process large amounts of data to forecast medical production and supply chain bottlenecks during future pandemics. The testing scenarios are built to tackle the logistical challenges and production disruptions and supply chains for vaccine distribution with optimisation algorithms	Research use	(Radanliev & De Roure, 2022)
	Novel hierarchical heuristic-simulation procedures	Decision support system that integrates GIS, analytics and simulation methods to help develop a priority-based distribution of COVID-19 vaccines in a large urban setting. The methodology applies novel hierarchical heuristic-simulation procedures to create a holistic algorithm for prioritising the process of demand allocation and optimising vaccine distribution	Research use	(Shahparvari et al., 2022)
	Stochastic optimisation model	Four-echelon supply chain to distribute plasma from newly recovered Covid -19 patients for treatment. It employs simulation to estimate plasma demand revealing increased costs and flow time with rising plasma demand	Research use	(Shirazi et al., 2021)
		Supply chain network with four echelons of manufacturers, hospitals, vaccination centres and volunteer vaccine students to enhance COVID-19 demand estimations, minimise supply chain costs, and maximise justice in vaccine distribution	Research use	(Kamran et al., 2023)
	Optimisation algorithm based on multiparametric disaggregation	Mathematical program to minimise the number of vaccine doses distributed to effectively extinguish an emerging outbreak in its early stages	Research use	(Enayati & Özaltın, 2020)

[a] *For this study the availability of the tool in the market was defined as: "Research use" for AI-based tools developed within the academia but not yet implemented or tested within a real-world setting, "In development" for AI-based tools currently under design or improvement. Overall, the methodology of these tools has been published in the literature*

because of their innovative approaches, "Open source" for AI-based tools for which the original source code is made freely available and can be redistributed and modified, and "On the market" for AI-based tools that have been already deployed in real-world settings and are currently being used.

2.3. AI supporting decision-making in health emergency preparedness and response

The main outcomes of the literature review on AI-based tools for supporting decision-making in health crisis are:

- AI-based tools can support **in identifying the best vaccination strategies** and in prioritising counter measures to be implemented.
- Reliable forecasts of epidemic outbreaks are essential for **predicting MCMs demand and adjusting production, stockpiling, and distribution**. ML and NLP algorithms analyse diverse data sources to help governments and health institutions optimize pandemic response strategies.
- AI systems drawing on different data sources can help **predict future pandemics and monitor current outbreaks** allowing the production and stockpiling of MCMs to be **planned** and pandemic health response strategies to be optimised.
- AI support in decision-making can **prevent the "bullwhip effect"** and support MCMs production and distribution according to real-time disease spread.

Evidence has shown that AI-based tools can support decision-making during a health crisis by providing insights, prediction and optimisation strategies regarding MCMs production, stockpiling and distribution by analysing diverse data sources and providing real-time assessments. Regarding the demand for MCMs, AI-based tools make it possible to predict future epidemic outbreaks in advance or monitor them in real-time through surveillance models, allowing the production and stockpiling of MCMs to be tuned and pandemic health response strategies to be optimised. AI-based tools that improve the diagnosis of new diseases can generate useful data for surveillance applications, improving their efficiency and effectiveness. On the supply side, AI-based tools can analyse data on the supply and availability of MCMs and healthcare infrastructure to optimise distribution strategies based on defined priorities. The following section describes the areas in which AI can contribute to supporting decision-making regarding MCMs production, stockpiling and distribution during health emergency preparedness.

Having reliable **forecasts of epidemic outbreaks** before or during a pandemic can be crucial in **predicting demand for MCMs** and adjusting their production, stockpiling and distribution accordingly. ML and natural language processing (NLP) algorithms can collect and analyse data from several sources such as news bulletins, social networks, airline ticketing, and reports related

to plant and animal diseases in different languages to inform the definition and optimisation of pandemic response strategies by governments and health institutions. An example of a tested application is **Bluedot**, an initiative dedicated to employing AI to detect outbreaks during epidemics promptly. Bluedot develops prediction models for virus detection which can help European and regional authorities to forecast MCMs demands given new outbreaks (Deo & Anjankar, 2023). For instance, it was reported that Bluedot detected the coronavirus outbreak in Wuhan before local authorities confirmed the first cases, showcasing its early warning capabilities. This tool aggregates data from diverse sources including social media, government documents, and healthcare organisation records, leveraging NLP and ML to build intelligence (Nguyen et al., 2021). A broader initiative called "Epidemic Intelligence from Open Sources Initiative" is driven by the European Centre for Disease Control and WHO and focuses on epidemic intelligence from open data sources (news, curated reporting systems from WHO, social media, mobile data to track citizens' movements) to identify new outbreaks of health threats by automatically extracting specific keywords from these sources. This tool is currently used by several institutions worldwide (WHO, 2019). Different methods, such as multi-layered perceptron and adaptive network-based fuzzy inference systems were also reported to be able to predict the COVID-19 outbreak effectively (Ardabili et al., 2020; Pinter et al., 2020). The refinement of these AI techniques and their use by governments and health authorities could ensure better preparedness for future pandemics, allowing the need to produce and stockpile MCMs in certain regions to be predicted in advance.

In addition to predicting future outbreaks, **AI can be useful in monitoring the actual spread of the disease across regions and countries through integrated AI-powered surveillance systems.** AI-based tools have been shown to enable the production, stockpiling and distribution of MCMs to be adjusted based on real-time data, minimising response time and waste, and thus maximising the efficiency and effectiveness of health emergency response. DL or ML approaches can be leveraged to establish an integrated model that combines different types of individual data models to build a surveillance system. This type of integrated model-based platform can contribute to identifying threats from infectious diseases of international concern as well as in infectious disease surveillance (Arora et al., 2021; Zeng et al., 2020). Another example is **SENTINEL**, which is a software system built upon recent developments in ML and data processing for real-time syndromic surveillance based on social media data. This system provides a fast and early assessment of disease severity and transmission potential to support decision-making in logistical planning in healthcare systems (Zeng et al., 2020). Another useful tool to support decision-making in estimating MCMs demand is an **AI model for predicting risk factors** based on travel history records along with common signs and symptoms through mobile phone-based online surveys (Srinivasa Rao & Vazquez, 2020). Simpler and more widely used during the COVID-19 pandemic are **contact tracing apps**, which allow tracking contacts between people, and facilitating the monitoring of the spread of the disease (Fontes et al., 2023). The EU toolbox on

contact tracing and warning apps emphasises that these apps should be voluntarily installed and used. **AI technologies can also help in detecting and diagnosing diseases with higher accuracy and at a faster pace.** Numerous AI companies have been actively engaged in developing tools for the efficient detection and diagnosis of diseases during the COVID-19 pandemic (Nguyen et al., 2021). Among them, **Infervision** responded to the COVID-19 pandemic by aiding clinicians in the efficient detection and monitoring of the disease. Infervision's technology was deployed at Tongji Hospital in Wuhan, China, at the epicentre of the outbreak. Their AI system has notably enhanced the speed of CT diagnosis, a crucial aspect for expediting the detection of pneumonia in suspected patients (Huang et al., 2023). Sticking to China, a group of physicians at Zhongnan Hospital in Wuhan utilised GPU-accelerated AI software, specifically **NVIDIA's** AI, originally designed for lung CT cancer detection, to recognise visual indications of the COVID-19 virus (H. Wang et al., 2023). In this regard, important steps have also been taken thanks to federated learning, which brings together data from different treatment centres worldwide. An example is the FL-AI project for COVID-19 chest CT scans which involved the collaboration of medical institutions from China, Italy, and Japan to improve the diagnosis of COVID-19 (Deng et al., 2022; Yang et al., 2021). Another example using federated learning is the multicentric system where each institution provides COVID-19 X-ray image datasets to train a **Global Adversarial Network**. The advanced federated learning scheme produces better synthetic images, leading to higher detection rates due to enriched feature learning and complete image construction across distributed institutions (Nguyen et al., 2021). Data from contact tracing apps, risk factor prediction models and AI-based tools to improve the detection and diagnosis of diseases can be integrated into surveillance systems to improve disease monitoring results, MCMs demand predictions and overall pandemic response strategies.

ML-powered decision support systems can also leverage social media analytics, geospatial analysis, and satellite imaging to forecast medical equipment logistic needs, predicting the regional risk classification colour codes, emergency room attendances, and the required regional medical supplies. A concrete example of this type of ML application is the **Earth Cognitive System 4 COVID-19** which can provide both predictive information, aiming to identify possible new clusters of outbreaks, and supervisorial information, by monitoring infrastructures, i.e. traffic jams, parking lots, or specific population groups, i.e. teenagers, doctors, teachers, etc. (Atek et al., 2023).

AI models have also proved useful in **optimising MCMs distribution and allocation strategies during health crises by assessing the current availability of MCMs and recovery structures** (Arora et al., 2021; Deo & Anjankar, 2023; Nguyen et al., 2021; Pall et al., 2023). Combined with applications to predict and monitor disease spreading and MCMs' demand, these types of models can streamline the allocation and distribution of MCM, and patients based on infrastructures and MCMs supply availability. By matching demand and supply-side data, several ML-powered

predictive decision support systems for pandemic response management and medical logistic planning were developed during the COVID-19 pandemic (Ardabili et al., 2020; W. Cao et al., 2022; Georgiadis & Georgiadis, 2021). An example is a mixed integer linear programming (MILP) model addressing the optimal planning of COVID-19 vaccine supply chains and the optimal planning of daily vaccinations in the available vaccination centres (Georgiadis & Georgiadis, 2021). Optimisation algorithms combined with vaccination allocation models can also determine near-optimal spatiotemporal allocation strategies for vaccines with varying effectiveness and coverage (W. Cao et al., 2022).

Moreover, **AI can support decision-makers in prioritising the measures to be implemented to counter pandemics**. ML models, and especially supervised ML models, can classify disease severity, attribute risk factors, **assign priority scores** to regions or population groups and combine this information to assess overall measures' priority (L. Cao, 2023). During the COVID-19 pandemic, ML models were used to define the best vaccination strategies, and many different models agreed on the recommendation of concurrent vaccination of the elderly population across different regions (Ali & Wood-Harper, 2022; L. Cao, 2023; Kamran et al., 2023). A concrete example is the decision support system for prioritised COVID-19 two-dosage vaccination allocation and distribution that integrated GIS, analytics, and simulation methods to help develop a priority-based distribution of COVID-19 vaccines in a large urban setting (Shahparvari et al., 2022). A more complex model is the stochastic multi-objective, multi-period, and multi-commodity simulation-optimisation model developed to support COVID-19 vaccine production, distribution, allocation, and stockpiling control decisions, setting up multiple objectives and priorities, such as minimise supply chain costs, maximise vaccination desirability, and maximise justice in vaccine distribution (Kamran et al., 2023).

Especially for this latter type of AI applications, but in general, for applications that aim to support decision-making, **governance aspects are crucial**, and it is always recommended to analyse and balance AI-based tools' outcomes with human intervention, by employing a human-in-command or human-in-the-loop approach (Anderson et al., n.d.).

Box 4. *Considerations from the experts: AI for decision-making support in health emergencies*

Highlights resulting from Experts' consultation

Experts from the industry and governmental sectors proposed **knowledge extraction models based on large language models** could be applied to most of the tools analysed in this review to notably improve decision-making in health crisis. To ensure **explainability and quality results**, one expert from governmental sector considered important to combine them with external retrieval (Retrieval Augmented Generation - RAG) so that information is taken from specific data sources and that **traceback is ensured**.

The expert mentioned that one of the main challenges of large language models are hallucinations that can take place for example when data is scarce or controversial and the models provide random answers. In this case, RAG can be used to mitigate the hallucination factor. The same expert mentioned that **Generative AI (GenAI) could be potentially used for causal learning** to identify the causes that originate epidemics or pandemics, such as climate causes. GenAI is one of the fast-growing sectors within AI industry and its application to early detection systems can be extremely useful to predict future pandemics or menage them while ongoing through effective surveillance systems. This capability could be boost by incorporating series social networks with GenAI.

In this area of supporting decision-making, experts also emphasised the need to establish collaboration and knowledge between scientists and healthcare organisations and professionals which was raised within the other areas covered in this report. Experts agreed that to realise the potential of AI for health emergency preparedness and response, it is **paramount to ensure smooth collaboration** and knowledge sharing between data scientists and healthcare professionals.

An expert from governmental sector but working in collaboration with the industry sector explained that AI has proven to be invaluable in various aspects of public health and pharmaceutical development. It supports decision-making models for developing new drugs, conducting pharmacovigilance, and MCMs. AI can assess in real-time the impact of MCMs on ongoing epidemics, **offering timely insights that are crucial for effective responses**. During the COVID-19 pandemic, AI was instrumental in tracking the emergence and behaviour of the pathogen, enabling health authorities to predict the arrival and spread of the virus by analysing social media data. This predictive capability allowed for more proactive and informed public health measures.
Starting from COVID-19 crisis, several programs to shorten reaction times were launched. Epidemiological surveillance was a big part of it, collecting and analysing data of how many patients have a specific disease, targeting specific actions to improve the reaction time when facing a new health crisis. Experts mainly from academia and governmental sectors emphasised the need to use surveillance and monitoring systems as well as creating an EU common system.

Other topics mentioned during the consultation activities are the following:

- The field of computer vision has been making great strides in recent times, especially in image recognition in the medical field. These advances can be applied on a large scale in the event of a future health emergency, to compare and recognise through computer vision the signs of emerging diseases, known cases and future cases.
- Large language models (LLMs) are used for decision support by practitioners to guide their diagnosis or more accurate use of drugs. This can be applied to emerging diseases in case of health emergency treat. In the realm of policymaking, LLMs can be used for understanding, verifying and contrasting the alarms that come up during a health crisis, to ensure their accuracy and consider all the contextual elements necessary to plan and set up an effective response.
- In the same way AI technologies and particularly DL models can be applied in weather forecasting, they can be very useful for medical forecasting. In medicine they can support health crisis preparedness with forecasting specific, actionable, high risk for individuals to prevent diseases or severe acute events. Multimodal AI can help in this by exploring several variable impacting risk factors. An example can be found in Alzheimer's disease where given the number of biomarkers and parameters that can contribute to the development of the disease it makes much more sense to not rely only on single biomarkers but include additional information like age and other variables to make a reliable prediction. These applications can help detect emerging diseases and their probability of impacting specific demographic groups of the population.

Finally, two experts, one from the industry and one from academia, highlighted the importance of digital twins in supporting decision-making. They can be employed to test various scenarios, make predictions, and anticipate events.

Additionally, generating digital twins can contribute to better informing the general population by creating tailored messages for different demographic groups. This ensures that the communication is more understandable and personalised to meet the specific needs of each group.

The experts mentioned the 'European Virtual Human Twins Initiative' initiated by the EC which intends to power up advanced supercomputing capacities and AI to facilitate collaborative virtual human twins research and technology development (European Commission, 2024). This initiative would support the development and simulations of models in ways previously not possible, speeding up the development of virtual humans' twins through enabling large AI foundational model training specific to human health and diseases.

Table 4 below provides a quick overview of the main AI-based tools identified to support decision-making during health crises, the AI technique utilised, and the level of maturity and implementation in the market.

***Table 4**. The main areas of AI implementation to support decision-making in health emergency preparedness and response, enabling technologies and AI-based tools identified through the literature review*

Area of implementation	Enabling technologies	Functioning and goals	Availability[a]	Author
Predicting pandemics and outbreaks	Multi-layered perceptron (MLP) and adaptive network-based fuzzy inference system (ANFIS)	Predict the COVID-19 outbreak as an alternative to susceptible–infected–recovered (SIR) and susceptible-exposed-infectious-removed (SEIR) models	On the market	(Ardabili et al., 2020)
Pandemics surveillance	Deep learning-based software	Aid in detection, localisation and quantification of COVID-19 pneumonia	Research use	(Zhang et al., 2020)
Disease transmission models	ML regression techniques	ML and complex networks for infectious disease transmission modelling	Research use	(Tripathi et al., 2020)
Risk factors prediction	ML	ML algorithm for risk factors prediction using a mobile phone-based web survey	Research use	(Srinivasa Rao & Vazquez, 2020)
Medical logistic planning	Regression modelling	Predictive decision support system for pandemics response management and medical logistic planning	Research use	(Atek et al., 2023)
	GIS, analytics and simulation models	Decision support system for prioritised vaccination allocation and distribution	Research use	(Shahparvari et al., 2022)
Detection and diagnosis of diseases	Heuristic vision for a public health large medical model	AI system for detection and monitoring of diseases as well as collect large-scale medical data in response to public health events	On the market	(Huang et al., 2023)
	Federated learning	Computer-aided characterisation and diagnosis with chest CT scan for virus identification	Research use	(Yang et al., 2021)
	Federated learning, Generative Adversarial Network	Generative Adversarial Network trained by different medical institutions to improve disease detection and diagnosis	Research use	(Nguyen et al., 2021)

[a] *For this study the availability of the tool in the market was defined as: "Research use" for AI-based tools developed within the academia but not yet implemented or tested within a real-world setting, "In development" for AI-based tools currently under design or improvement. Overall, the methodology of these tools has been published in the literature because of their innovative approaches, "Open source" for AI-based tools for which the original source code is made freely available and can be redistributed and modified, and "On the market" for AI-based tools that have been already deployed in real-world settings and are currently being used.*

2.4. Advantages and disadvantages of AI-based tools aimed at improving MCMs production, stockpiling, distribution and decision-making for crisis preparedness and response

The main advantages and disadvantages of the different AI technologies, techniques and technical approaches powering the identified tools within the literature review are summarised in **Table 5**.

***Table 5**. Advantages and disadvantages of enabling technologies for MCMs production, stockpiling, distribution, and decision-making during health emergencies*

	Advantages	Disadvantages
Machine Learning and large language models	• Data accuracy • Diverse applications of data use • High data security • Fast processing capacity, automation • Reduce human bias	• Data dependency • Hight cost of implementation • Data quality could be compromised by incomplete or inconsistent data • Generalisability of results to wider populations (re-calibration is needed when implementing ML across settings). Representativeness. • Bias and fairness (ML algorithms learn from existing data and therefore inherent biases in the training data can lead to biased outcomes)
Blockchain	• Data transparency • Data confidentiality • Reliability (reduces the possibility of data tampering) • Fast processing capacity	• Hight cost of implementation • Data privacy could be impacted due to cybersecurity issues • There are no guidelines or standards to guide implementation • Interoperability with existing IT infrastructure
IoT and smart sensors	• Personalised used experience • Real-time data processing and analysis • Advanced security and risk management • Autonomy in the connection	• Complexity of connecting IoT infrastructure and data with streams with AI algorithms • Regulatory and legal concerns due to data ownership generated by IoT devices • Data security (data are retrieved and transmitted from various sources) • Bias and fairness (ML algorithms learn from existing data and therefore inherent biases in the training data can lead to biased outcomes)

2.5. Bottlenecks and barriers to the implementation of AI solutions for MCMs production, stockpiling, distribution and decision-making in health emergency preparedness and response in the EU

This section outlines the bottlenecks identified through both literature review and consultation activities. The bottlenecks have been prioritised based on expert opinions gathered during the

consultation activities and based on their impact on the application of AI-based solutions for MCMs production, storage, distribution and decision-making in health emergency preparedness and response. The most pressing bottlenecks are described below, along with potential solutions to overcome them.

1. Data availability, quality and interoperability

Description

- Countries and the industry are reluctant to share their databases, limiting the ability of international health organizations such as the WHO to assess epidemics and respond effectively to health threats comprehensively.
- Healthcare professionals typically possess demand-side data, which enables them to identify shortages and anticipate patient needs. However, they often lack access to supply-side data, including information on stockpiling, which limits their ability to make informed decisions. Insufficient data on MCMs supply chain restricts the ability to optimise production scale-up, stockpiling, and distribution, resulting in potential stockouts, overstocking, or inefficient allocation of resources.
- Data available are scattered across the medical units of the countries and there is no unified system to report data, to ensure data quality or to integrate different data sources avoiding biases. One key requirement to ensure successful AI implementation for MCMs production scale-up, stockpiling and distribution requirement would be to have FAIR data (findable, accessible, interoperable and reusable data) across MCMs supply chains.
- The lack of a legal basis for collecting data on MCMs demands, stockpiling and distributions chains across countries presents a significant limitation in developing effective AI tools.

Proposed solutions

- Establish a legal framework for data collection and processing that encourages companies and institutions to share and analyse data on MCM demand, stockpiling, and distribution, while ensuring compliance with privacy and security standards
- To encourage private companies to report their data and envision new ways to involve them by ensuring disclosed data will only be used for the general interest of the population and will not be disclosed to any third party. A working precedent of how clear data governance rules might incentivise private companies to contribute to research is the High-Performance Computing (HPC) initiative to pre-check medical measures against COVID-19 put in place at the beginning of the pandemics with significant contribution of data by industry**.** Experts suggest that this solution requires a legal framework to ensure companies provide data and prevent information bias that may arise from sharing only partial data as well as developing pilot vulnerability assessments.
- Implement a comprehensive system for collecting data from various medical sources, ensuring that the data is reformatted into a common format to maximise the automation of data collection and analysis.
- To implement hybrid data encryption to secure data in sources (medical units' systems), and while transferring and storing on destinations (centralised or decentralised servers) using known cryptography methods and blockchain.

Sources: Arora G. et al. 2021; Pall R. et al. 2023; Radanliev P. and De Roure D. 2023; Kamran M.A. et al. 2023; Goodarzian F. et al. 2022.

2. Lack of standardised databases

Description

- Lack of standardisation in reporting of health data across Member States and lack of standardisation agencies within EU.
- The absence of standardised and integrated datasets for communicable diseases, including information on infected areas and the need for MCMs, hinders data interoperability that can lead to unreliable results and delayed actions.

- Lack of a standardised system to collect and process data into databases is a challenge.

Proposed solutions

- Development and implementation of a new legal framework, along with the need for MS, to redesign or rebuild existing databases and systems to integrate data into the new system. This would also involve duplicating or reusing existing systems within the EU.
- To build a comprehensive system that collects health data from various medical units, and securely stores the data in each country. In this way, a standard data storage format will be defined for unification and ready for sharing. This system would be useful not only during health emergencies but to support decision-making and other fields such as the Critical Medicines Alliance.
- To implement a decentralised system by building a data-sharing platform for the EU to connect to the country's medical data servers and collect their data during a pandemic or for research.
- To define data access for transparency and analysing security during data transfer and storing.
- Another possible solution would be that of adopting a certification system for new medical devices (not medicines) under a model like the US framework. The US has a certification model based on a network, composed of healthcare professionals and universities, that pre-certifies new devices (for example masks, respirators). This network acts as a precertification institution: if the network reports good quality of a new product the national competent authority is quicker at approving the product.

Sources: Nguyen D.C. et al. 2021; Shirazi H. et al. 2021.

3. Limited testing of AI for health emergency preparedness and response

Description

- The use and utility of AI for health emergency preparedness and response remain largely untested and it is usually relegated to a limited scenario testing.

Proposed solutions

- To conduct targeted scenario-based and digital twins testing to evaluate the performance and utility of AI systems in simulated emergencies.
- To allocate dedicated funding for testing and evaluating tools and scenarios.
- To foster collaboration and knowledge sharing between public organisations and the private sector.

Sources: Arora G. et al. 2021; Radanliev P. and De Roure D. 2023; Sarfraz A. et al. 2022

4. Validation and collaboration

Description

- AI validation techniques and gold standards are not sufficiently clear because these technologies are relatively new and evolving, and there has been no opportunity to use the same validation methodology several times.
- Robust validation of newly developed AI-based solution is needed, especially in the health sector, including sensitivity analysis or robust statistics, gain experience and establish gold standards. To do so, domain expertise in health emergency management needs to be merged with technical and AI expertise.

Proposed solutions

- To establish gold standards for health emergency management newly developed AI-based tools validation
- To put in place stable communication channels among domain experts and AI/technical experts

This would require guidance from regulators like EMA/FDA to allow companies to share this data in a safe, controlled way.

Sources: Chalmeta R. et al. 2023, Shah U. e tal. 2022, Kamran M.A. et al. 2023

5. Computational resources requirements

Description

- Deep-learning and blockchain solutions might require substantial computation resources to be deployed in an effective manner, implying high implementation costs. Similarly, IoT and robotics might imply high implementation costs in terms of infrastructure and hardware components

Proposed solutions

- Optimise computational resources and hardware requirements
- Provide adequate funding

Sources: Shirazi H. et al. 2021; Georgiadis G.P. and Georgiadis M.C. 2021; Deo N. and Anjankar A. 2023; Nguyen D.C. et al. 2021.

6. High energy consumption

Description

- Some more sophisticated AI-powered applications might imply high energy consumption, rising implementation and sustainability concerns (e.g. extensive energy consumption might be required to run consensus processes in the blockchain).

Proposed solutions

- Optimise energy consumption of developed AI applications

Sources: Nguyen D.C. et al. 2021; Twaróg S. et al. 2021; Shayannia S.A. 2023.

7. Data storage and data governance

Description

- The volume of data generated from AI devices creates diverse challenges in developing data strategies for training AI algorithms that can operate with lower memory requirements.
- Designing sparse compact and efficient AI algorithms for complex linked healthcare systems demands prior data strategy optimisation and decision-making on collecting and assessment of training data

Proposed solutions

- Data strategy optimisation before or during algorithms development

Sources: Radanliev P. and De Roure D. 2023.

8. Prioritisation ethical issues

Description

- AI-powered applications helping in decision-making for health crisis management could rise several ethical and prioritisation issues. For example, a vaccine allocation algorithm could suggest which regions, groups or individuals to prioritise for vaccine administration. An error in the calculation of this type of algorithm could delay the supply of vaccines or medicines to those who need them most

Proposed solutions

- Careful and iterated testing
- Risk management
- Human supervision

Sources: Arora G. et al. 2021; Cao L. 2023; Kamran M.A. et al. 2023; Sarfraz A. et al. 2022; Atek S. 2023; Shahparvari S. et al. 2022; Cao W. Et al. 2022

9. Regulatory considerations

Description

- AI can pose a legal and regulatory challenge if there is no party that is responsible and can be held accountable.
- Other relevant legal issues concern collecting information, content and personal information running on blockchain and AI platforms, such as problems with copyright infringement and defamation.

Proposed solutions

- Compliance with pharmaceutical regulations such as GMP, Good distribution practice (GDP) and Good clinical Practice (GCP).
- Defining legal schemes and internal governance models to dictate the governing law for AI operations
- To implement AI systems based on the EU Trustworthy AI guidelines and AI Act
- To promote accessibility and leverage AI regulatory sandboxes
- To collect data based on General Data Protection Regulations (GDPR)
- To prevent copyright issues, using open-source tools with permissive licenses or creating novel systems from scratch based on dedicated support and funding is possible.

Sources: Nguyen D.C. et al. 2021; Shayannia S.A. 2023.

10. Privacy

Description

- Ensuring the protection of individuals' privacy is of utmost importance in all AI-based solutions and applications. Governments may use mobile location data to track health treats, but it is imperative that this approach safeguards user data privacy, particularly sensitive information like home addresses, banking details, and shopping records

Proposed solutions

- Data anonymisation and encryption
- Securing information through blockchain and protecting privacy through roles management
- Legal measures and enforcement by relevant authorities
- To use data sampling by generating sample data with AI for emergency drills to ensure preparedness without compromising real data.

Sources: Ai Y. et al. 2022; Arora G. et al. 2021; Nguyen D.C. et al. 2021

11. Security

Description

- Data collected from patients, clinical labs and hospitals can be modified by data threats. Malicious attacks can inject false data or adversarial sample inputs which makes AI learning invalid, while the AI model can be tampered with. Therefore, security issues should be given high priority during the design and development stages.

Proposed solutions

- To leverage state-of-the-art cybersecurity tools for system protection and evaluation.
- To employ AI-powered tools for anomaly detection in systems, network traffic, and data.

- To utilise penetration testing and digital forensics expertise to assess vulnerabilities and penetrations, through dedicated specialist teams or outsourced services

Sources: Nguyen D.C. et al. 2021; Radanliev P. and De Roure D. 2023; Radanliev P. and De Roure D. 2022; Ai Y et al. 2022; Gomasta S.S. 2023.

12. User acceptance and trust

Description

- As for every innovation, user acceptance should not be given for granted. Given security, liability and privacy risks posed by AI applications for health crisis management, trust can be built upon transparency and clear communication, accompanied by robust security measures, clear accountability, and strong data governance policies.

Proposed solutions

- Trustworthy AI development
- Collect users' feedback
- Clear communication to users
- Accessible communication channels for citizens to complain
- Watermarking of AI generated content

Sources: Deo N. and Anjankar A. 2023; Bas T.G. et al. 2023.

13. AI literacy

Description

- Healthcare professionals might be lacking the required skills to operate and manage AI-powered tools in hospitals, pharmacies, and other healthcare hubs.
- AI tool training is required not only in clinical areas but also within pharmaceutical manufacturing and supply chains, as well as among governments and regulators, to ensure an understanding of the capabilities and limitations of the new tools being provided, and how they can be effectively implemented.

Proposed solutions

- Providing dedicated AI training to healthcare professionals

Sources: Bas T.G. 2023; Deo N. and Anjankar A. 2023.

2.5.1. Future areas of research

Given the bottlenecks and barriers identified above, key areas for future research highlighted within the literature include **piloting and testing** across the four research areas analysed in this study and **improving the security** frameworks for AI systems in health crisis management. Regarding piloting and testing, existing studies mostly address a single epidemiological landscape, while the robustness and effectiveness of the proposed strategies should be tested under other scenarios and landscapes (L. Cao, 2023). On security, future research should focus on enhancing security to ensure systems are safe, secure, and transparent, and operate within safe environments while adhering to legal and ethical standards. In terms of cybersecurity,

researchers emphasise future systems need to be traceable and explainable, responding to threats according to the principle of proportionality and resilient against AI-powered cyber-attacks. Research should also prioritise training AI algorithms to prevent both active and passive recognition by adversarial AI on specific targets at scale (Radanliev & De Roure, 2022). A substantial effort is required to build unified databases and templates to collect data on the MCMs production scale-up, distribution and stockpiling across the EU countries and regions (Nguyen et al., 2021). In addition, as mentioned above, an additional underlying challenge is the absence of a regulatory framework to approve these tools for pharmaceutical use. To begin developing this framework, first, the EMA or other national agencies need to gain a comprehensive understanding of the AI tools and their applications and then, establish testing environments and outline a roadmap with criteria for full approval.

For stockpiling, distribution and routing algorithms, the existing literature agrees that efforts should be directed towards broadening the scope of the applications and working towards interoperability and convergence. This includes further exploring the suitability of AVs and UAVs for delivering emergency MCMs and their possible integration in established transportation routes and warehouse networks. Furthermore, there is room to develop additional metrics for evaluating generated routes, considering factors such as energy consumption or the risk of closure due to hazards (McDaniel et al., 2023). Other goals mentioned in the studies are the expansion of digital twins to develop vaccines and drugs, simulate disruption impacts, test alternative distribution options and support decision-making during health emergencies by using real-time data (Kamran et al., 2023), while also helping decision-makers understand where AI has positively influenced their choices. Some of these processes are still in the early stages, with only limited companies involved, indicating that full industry-wide adoption has yet to occur. However, advancements are gradually progressing. For example, patient twins and personalised health have been proposed as key technologies to ensure vaccines and treatments work given the individual's markers (Popa et al., 2021; Sahal et al., 2022).

Finally, strategies based on the adoption of AI-based tools have proven to be useful in fighting COVID-19, but the potential of these strategies has yet to be identified for future pandemics and health crises in the four areas covered in this report. It is essential to discuss further ethical aspects related to the functioning and impacts of AI as well as its governance at the institutional level and through bottom-up approaches (Fontes et al., 2023). Legal frameworks have been proposed at the EU level and apply as well to MCM-related solutions based on AI. Thus, research should also focus on identifying legal gaps and uncertainties that can undermine the safe implementation of such systems. At the technical level, infrastructure capacity, computational power and data repositories will play a critical role in supporting the implementation of such systems and need to be reinforced/expanded (Arora et al., 2021).

3. Discussion and conclusions

In this study, we aimed to explore the use of AI-based tools to improve MCMs production, stockpiling, distribution, and decision-making for emergency preparedness and response. Through an extensive review of the literature and consultation activities, we investigated the current state of the art, identified bottlenecks, and discussed their potential impact on health preparedness and response.

In terms of the use of AI-based tools for MCMs production, stockpiling and distribution, this study highlights the growing importance of AI in addressing **digitisation and improving the performance of the MCMs production, stockpiling and distribution processes**, as it has been observed that the AI has the potential to improve productivity and efficiency by leveraging digital manufacturing technologies such as AI and IoT. Another interesting point raised by this review is the need to **improve connectivity and collaboration between healthcare organisations, industry and policymakers** to improve the response to MCMs demands, prevent shortages and improve overall MCMs distribution efficiency. The collaboration between stakeholders should aim at sharing and using real-time data from suppliers, inventory levels, production schedules, and customer demand.

Existing tools and AI techniques have been mapped according to the targeted areas within the supply chain (production scale-up, stockpiling, distribution) as well as supporting decision-making. **The results show how AI-based tools can contribute to enhancing production scale-up, especially for vaccines and identifying new molecules to improve existing treatments.** In addition, AI can foster cell lines providing insights into potential targeted therapies advancing in personalised medicines and increasing the efficiency of biopharmaceutical processes. In terms of quality control, by leveraging IoT devices and ML algorithms real-time data from all production lines can be collected. Another example of where AI could play a significant role is in the implementation of Process Analytical Technology systems. These systems use modern analytical techniques to monitor Critical Quality Attributes of products in real-time, enhancing the understanding and control of the production process. By doing so, they can enable immediate adjustments, minimise losses, and improve overall efficiency. Overall, thanks to AI-based tools, the development and usage of pharmaceuticals are expected to accelerate in the coming years because of recent technological advancements in data gathering and generation tools, reliable information management and exchange platforms, and enhanced computational capabilities.

Overall, evidence has shown that using AI for MCMs stockpiling and distribution in health crises can greatly enhance efficiency and effectiveness by **improving the allocation of MCMs** through **optimisation of MCMs allocation and delivery processes** according to real-time demands, and **automatisation of inventories** and orders. These techniques can leverage data for identifying the **best MCMs distribution strategies** that prevent MCMs shortages, detect sales anomalies and enhance warehouse automation to ensure fairness in accessing MCMs**.** In addition, AI-enabled models have proven to be useful in leveraging existing product and process knowledge

to swiftly determine the best processing parameters, cutting down on design time and reducing waste.

In terms of AI-based tools aimed at supporting decision-making in the above-mentioned areas, evidence shows they can **furnish decision-makers with more precise information** about public health events, thereby **enhancing strategic planning and next-response efforts**, which is particularly crucial in addressing health emergencies. Consequently, through AI-based tools and systems, decision-makers can promptly access information on disease transmission, health risks, and allocation of MCMs, enabling them to formulate more effective public health policies and action plans. In addition, the results of this literature review also show how AI-based tools can support authorities from Europe down to the local level to identify the best strategies for MCMs production, stockpiling and distribution to enhance effectiveness and rapid response when facing a health emergency. The use of AI-based tools in the above-mentioned areas overcomes previous limitations as AI allows the analysis of vast amounts of data to better predict MCMs needs as well as the interconnection of all the systems of all the actors involved in MCMs production, stockpiling and distribution.

Despite significant advancements, this review has identified barriers and bottlenecks to the implementation of AI-based tools in the fields tackled by this review. Overcoming these bottlenecks remains a priority for the involved stakeholders to allow a successful use of AI within the MCMs production, stockpiling, distribution and decision-making. Particularly, the main bottlenecks that emerged were related to **data availability, quality, and interoperability** since ensuring data quality is key to eliminating biases in medical databases and addressing the unfairness of AI medical algorithms. Recently, widespread concerns have emerged about AI-based tools due to insufficient data diversity and algorithmic bias and the need to mitigate them in the future of AI-based tools as indicated by academic developers and governmental stakeholders. These regulatory hurdles are currently being discussed by key national and European agencies and organizations such as the EMA, FDA or the European Consortium for Continuous Pharmaceutical Manufacturing (ECCPM) to explore the feasibility of implementing a new regulatory framework (Arden et al., 2021). In addition, regarding integrating AI into drug development processes careful attention to regulatory considerations, emphasising data privacy, consent, and responsible handling of patient data. Additionally, as mentioned before, implementing **collaborative networks** between stakeholders is crucial for the successful integration of AI within the healthcare supply chain.

In line with the above, the analysis of the availability has revealed that many of the identified AI-based tools have been used for research purposes but still require further development before they can be deployed in the EU market indicating the need for further development, validation and assessment of their external generalisability for an effective implementation in the European context (Annex 2). Nevertheless, these concerns and bottlenecks are expected to be overcome in the coming years and more AI-based solutions are expected to be implemented in the EU to

enhance the potential of AI in the areas of MCMs production, stockpiling, distribution, and decision-making.

This report highlights that the ability to integrate all the available data into AI models is still at an early stage in the EU, and until now has been mainly restricted to electronic health records and medical images due to data availability and interoperability limitations. In addition, personal data or continuous data from wearable sensors have yet to be incorporated, as well as other data sources, particularly from automated warehouses and supply chains.

Finally, drawing on the significant advancements in AI over the past few years and the progress made in other fields presents an opportunity to greatly enhance MCMs production, stockpiling and distribution processes to improve resilience and prevent vulnerabilities during future health emergencies.

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Annex 1. Methodology

This section outlines the methodology followed in this report for the review of the literature on the use of AI-based tools for improving MCMs production, stockpiling, distribution and decision-making. Moreover, this section also describes the methodology used to gather additional inputs on the topic from key relevant stakeholders, covering from the selection of relevant stakeholders up to the analysis of the inputs gathered.

Literature review

The systematic literature review was intended to analyse the literature referring to AI-based tools and systems applied to production scale-up, stockpiling, distribution and decision-making in health crisis preparedness and response. An extensive literature search was conducted following the PRISMA guidelines and using the academic databases PubMed, IEEE and ACM Digital Library. **Box 5** below presents the search string used for the systematic literature review, which was developed as a result of several discussions among the team involved in the study and tested to ensure valid and reliable outcomes were obtained. Studies published within the last 10 years (2015-2024) were considered, and all eligible studies had to be open source, the full text available, and be written in English.

Box 5. *Query string.*

Titles and abstracts of articles published between 2015 and 2024 were searched for the following terms in PubMed:

("artificial intelligence"[Title/Abstract] OR "AI"[Title/Abstract] OR "algorithm*"[Title/Abstract] OR "ML"[Title/Abstract] OR "Machine Learning"[Title/Abstract] OR "Deep learning"[Title/Abstract] or "LLM"[Title/Abstract] or "large language models"[Title/Abstract]) AND (“production scale-up”[Title/Abstract] OR “manufacturing efficienc*”[Title/Abstract] OR “supply chain”[Title/Abstract] OR “supply network*”[Title/Abstract] OR “emergency planning”[Title/Abstract] OR “warehouse management”[Title/Abstract] OR “inventory management”[Title/Abstract] OR “logistics”[Title/Abstract] OR “stockpile”[Title/Abstract] OR “dispensation”[Title/Abstract]) AND (“medical countermeasure*”[Title/Abstract] OR “mcm”[Title/Abstract] OR “medical counter measure*”[Title/Abstract] OR “pharmaceutical” [Title/Abstract] OR “Drug” [Title/Abstract] OR “Vaccine” [Title/Abstract])

Search results were downloaded, resulting in one file prepared for the screening of the search hits (n=109). References were screened by title and abstract to identify whether the studies mentioned AI-based tools or systems (or other emerging autonomous technologies) that are or could be used to 1) improve the different areas across the production of MCMs during a health crisis, 2) increase efficiency or effectiveness of MCMs stockpiling and distribution during a health emergency, 3) support decision-making on MCMs demands and access during health emergency preparedness and response, 4) identify the most important bottlenecks to the implementation of

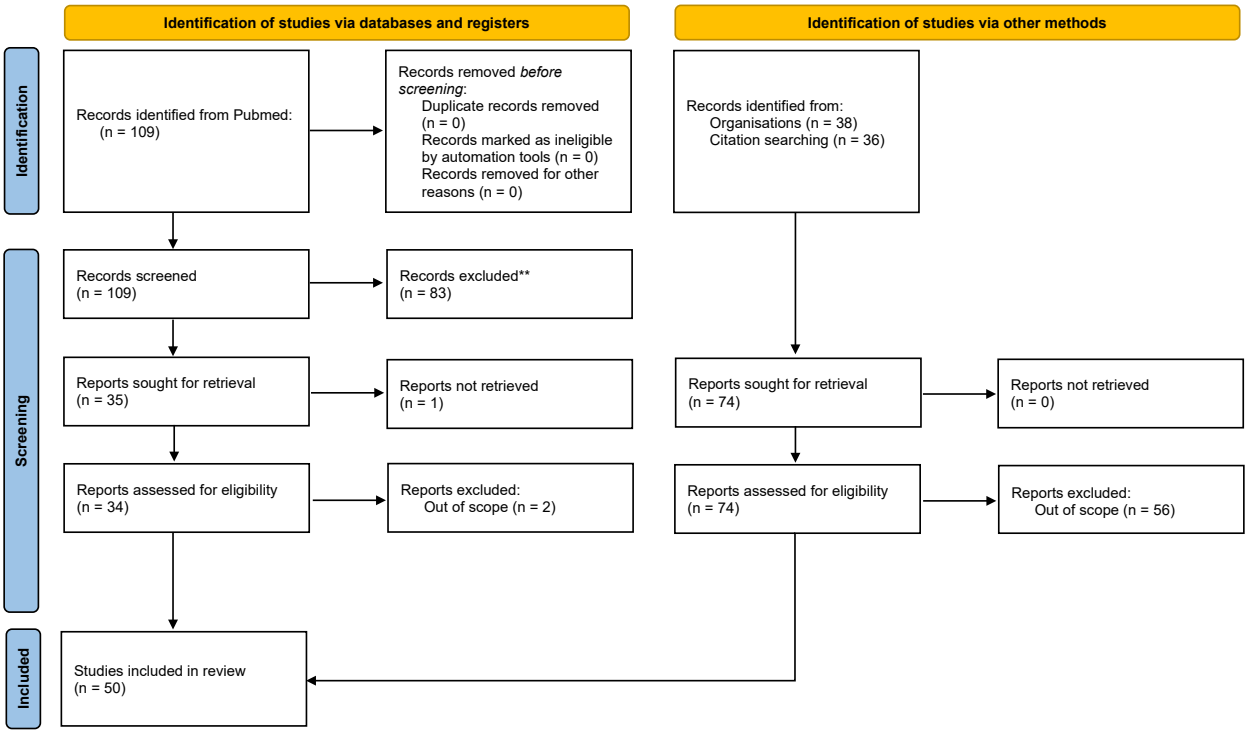
AI solutions in the areas as mentioned earlier and 5) describe the availability of the identified AI tools within the literature. A total of 35 publications were selected for further full-text review.

A full-text review was conducted before potentially eligible reports were further assessed. Studies were excluded if they were: 1) not peer-reviewed or from a reputable institution, 2) outside of the research scope, and 3) non-relevant to the research questions. A total of 32 studies were selected for data extraction. Reasons for exclusion for the other studies were recorded. The study team manually added 18 key references identified outside of the database search, which emerged from checking the references of relevant publications and was considered key (snowballing approach). A single reviewer extracted data from each included report (n=50). A second reviewer checked extractions and appraisals for accuracy and completeness. A narrative synthesis was performed.

The total number of records after each screening round was documented using the PRISMA flow diagram template (Error! Reference source not found.)(*PRISMA Flow Diagram*, 2024). The review of the literature allowed the identification of some authors that explored AI-based tools focused on MCMs production scale-up, stockpiling and distribution, analysed how these tools can contribute to improving the mentioned areas, and presented different options and strategies to prevent MCMs shortages and better distribution and stockpiling systems.

Note: This report identifies AI-based tools primarily focused on health emergency preparedness and response. While other AI-based tools could be applicable in this field, the report specifically highlights those designed to address MCMs production, stockpiling, distribution, and decision-making in the first instance.

Figure 1. *PRISMA flow diagram summarising the records of the literature review. From Page et al. (2021)*



Consultation activities

The consultation activities were carried out for two months and aimed to achieve three primary objectives:

- **Validate and expand upon the research gaps** identified in the literature by incorporating the experiences and perspectives of experts from academia, industry, and government.
- **Identify and prioritise the most crucial bottlenecks** and obstacles to the adoption of AI in crisis preparedness and emergency response.
- **Assess the availability and maturity level** of the identified AI-based tools to support on the implementation of AI-based tools within the EU.

To meet these objectives, the consultations included in-depth one-on-one interviews (60 minutes long) and a structured survey. Before the consultations, three distinct interview guidelines were developed, each tailored to a specific stakeholder category (academia, public health laboratory, industry, and government). These guidelines were reviewed and validated by HERA, ensuring comprehensive coverage of key topics while allowing flexibility for individual expertise.

Each guideline included the following sections:

- Introduction and consent to recording and minutes
- General questions about the interviewee's profile and expertise in the field

- questions related to the AI tools in the 4 main research areas (MCMs production scale-up, stockpiling, distribution, and decision-making in health crisis preparedness and response)
- Questions related to research bottlenecks and evaluation of tools maturity, and
- Questions about additional sources or experts to consult.

A comprehensive stakeholder mapping exercise was conducted at the outset of the study (and lasted six months), identifying key experts across the four stakeholder categories. This mapping was validated by HERA and informed the subsequent engagement activities, which included targeted emails to request participation in interviews and the survey.

The consultation strategy envisioned a target number of participants for each category:

- Academia (N = 5)
- Industry (N = 5)
- Government (N = 5)

During the two-month consultation period, other stakeholders from those originally selected were included in the engagement activities, after a thorough review of their expertise. **Table 6** below summarises the number of stakeholders contacted for each consultation activity and the rate of positive answer.

Table 6. *Inclusion of stakeholders in the consultation activities*

	Survey (N)	Interview (N)	Positive answer (%)
Academia	49	16	5%
Industry	12	6	28%
Government	12	7	21%

Ultimately, **three** stakeholders from academia, **five** from industry, and **four** from government were consulted, providing invaluable feedback and input.

The interviews were conducted in a semi-structured manner, leaving room for further tailoring to the interviewee's expertise. Approximately one week before the interview, some preparatory materials were shared with the interviewees, containing preliminary tables of the AI tools identified by the study and research bottlenecks. Following the interview, the minutes of the meeting were shared with the interviewees for review and approval, before incorporating the input and feedback into the present study.

As for the survey, it was conveyed through the EUSurvey tool and was distributed to all experts mapped in the three stakeholder categories not selected for peer reviewing or interviews. The

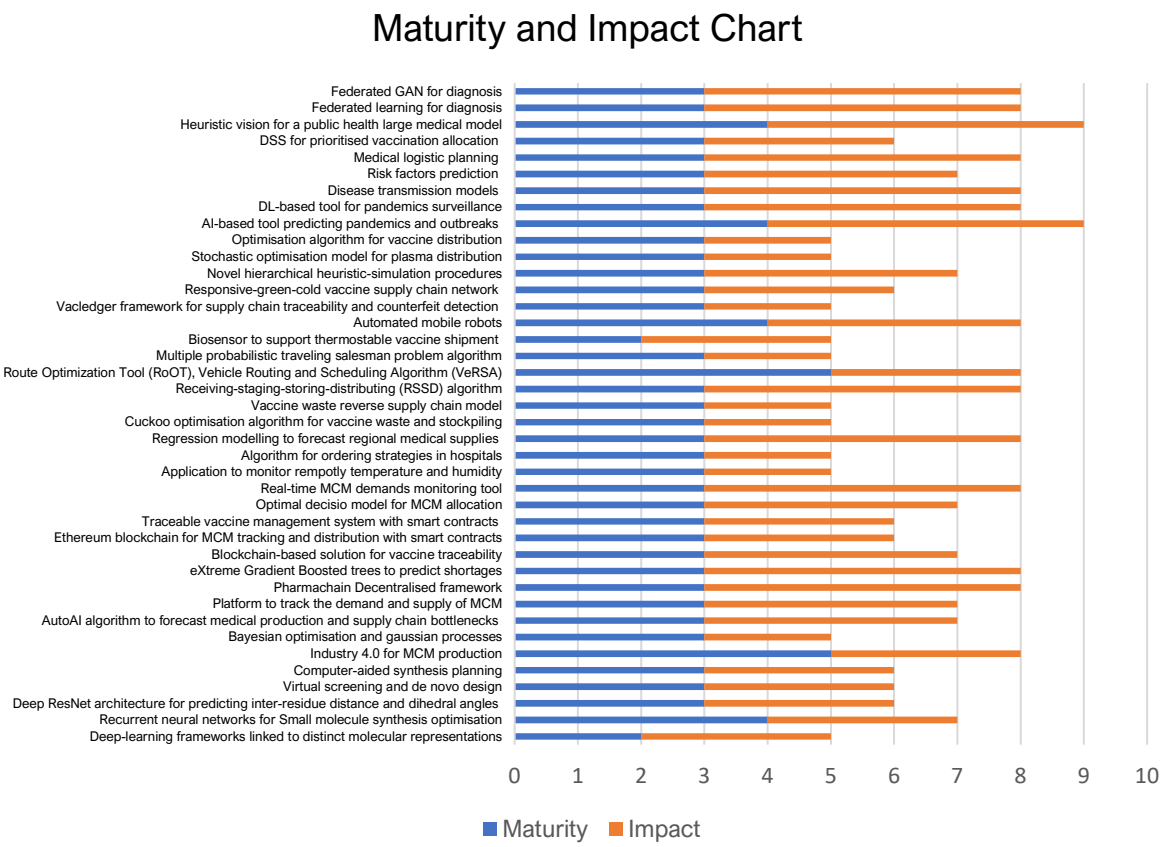
survey was structured in the same way as the interview guide, containing a first section of general questions, a second on AI tools in the four research areas and a section on research bottlenecks.

The valuable insights gathered from the consultation activities are presented at the end of each relevant results section (Box 3, Box 7, Box 9). These findings collectively contributed to the final version of the research bottleneck table in section 2.5.

This scientific background document has been peer-reviewed by three experts, and their comments and suggestions were incorporated into the final version.

Annex 2. Analysis of maturity and impact of the AI-based identified tools

Figure 2. *Maturity and impact of the AI-based identified tools*



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